

THE EFFECTS OF PRIVACY REGULATION ON THE MARKET FOR STOLEN DATA

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Abstract

Organizations collecting data on individuals are frequent victims of data breaches. Online stolen data markets have formed to facilitate trade between those who steal data and those who wish to exploit it. I propose a model of the stolen data economy and show that a privacy regulation decreases the expected number of bundles of stolen data available while also alleviating the market's asymmetric information problem. The latter effect increases breach probability for the most data-rich organizations, potentially increasing expected breach size. I then introduce a novel dataset of data breaches to measure the effect of the European Union's General Data Protection Regulation (GDPR) on the quantity of data available in the illicit market. Using a difference-in-differences design, I find that the GDPR caused a 60 percent reduction in the number of data breaches posted, but no statistically significant change in the aggregate amount of data available. At the individual breach level, I find a nearly 70 percent long-run increase in the amount of data they contain. These results are consistent with the model's prediction of a shift towards more data-rich targets post-regulation due to a decrease in the equilibrium level of scammers in the stolen data market. (JEL: D82 K24, L51, L86)

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I. Introduction

When individuals interact with businesses, schools, and almost any other modern organization, they generate streams of data containing their names, financial information, addresses, religious and political views, and more. While producing all of this information has the presumed benefit of allowing the organizations collecting it to provide better services or more relevant advertising, it also puts data subjects at risk of that data being improperly accessed and misused. In 2023 alone the average digital identity appeared in nine separate data breaches and over one billion emails and passwords could be found online (SpyCloud, [2024](#)).

Exposed data is a valuable commodity for cyber criminals. It can be used to commit identity theft and fraud, and as the starting ground for future data breaches. Online markets for the trade of stolen data have developed where bundles of data are swapped for money, reputation, and bragging rights. Trades are conducted in Telegram channels, on the dark web, and niche forums on the clear web.¹ These markets allow even those lacking in technical skills to gain access to stolen data with relatively little effort, but they suffer from asymmetric information problems. Scammers will frequently offer goods that are fake or never delivered, driving prices down (Herley and Florêncio, [2010](#)).

In this paper, I explore how data privacy regulation affects the market for stolen data. I first propose a model of the stolen data economy to show that, although privacy regulation will likely decrease the expected number of data breaches, it may not reduce the amount of stolen data available in the market. In fact, such a regulation could even increase the amount of data available in the market. This is because privacy regulations reduce the equilibrium level of fraudulent data sellers, alleviating the market’s asymmetric information problem. I then introduce a novel dataset to estimate how the European Union’s General

¹The dark web is the portion of the web that is intentionally obfuscated and only accessible through specialized internet browsers. The clear web consists of websites that can be reached by anyone and will be indexed by search engines. Clear web forums that facilitate the trade of stolen data typically require a user creating an account to view and participate in the market, technically making them part of the deep web.

Data Protection Regulation (GDPR), one of the most comprehensive data privacy laws in the world, changed aggregate and product-level outcomes in the stolen data market.

The GDPR is a broad-reaching regulation that governs the collection and processing of personal data by covered organizations. It explicitly states under what circumstances data collection is considered lawful, and prohibits the processing of sensitive data with few exceptions. Additionally, the GDPR gives individuals the right to have their data deleted, transferred, or rectified; requires detailed record keeping on data collection, impact assessments prior to data processing, and the designation of a Data Protection Officer; and increases cybersecurity investment requirements. Notification requirements and large fines also significantly increase the cost of suffering a data breach. Demirer et al. (2024) estimate the GDPR increased the cost of data storage by 20 percent on average, and caused a 26 percent decrease in data storage among firms in the EU relative to comparable American organizations.

By reducing the amount of data collected and requiring increased cybersecurity, the GDPR reduced the availability of the primary input to illicit data markets: data. Effectively, the GDPR is a negative supply shock for data thieves. Breaking down the stolen data supply chain into two parts—legal data collection and data theft—makes this effect clear.

Legal data collectors gather information on their customers, employees, and users for marketing and general operations. These agents are directly regulated by the GDPR. They change their data collection and security practices to comply with the law.

After its legal collection, data is stolen by cyber criminals. For hackers, each potential target has an expected value and cost of hacking. They will choose how to allocate their efforts across targets based on these expectations and the conditions of the market. By reducing the amount of data collected and requiring organizations to invest in security, privacy regulations should decrease the value and increase the cost of breaching regulated organizations. This changes the set of targets worth hacking, and the supply available in the market.

In stolen data markets, participants are at least semi-anonymous, and only sellers can

fully observe the true value of their goods prior to purchase. This creates significant risk of adverse selection, making buyers willing to pay less for data than it is actually worth (Akerlof, 1970; Herley and Florêncio, 2010).

The key theoretical finding of the model is that, under the right conditions, even if privacy regulations like the GDPR reduce the number of data breaches, they may not reduce the amount of data stolen. This is because the reduction in breach probability is concentrated among mid-sized data collectors, while the most data-intensive organizations see their breach probability increase. The driving force behind the latter effect is a decrease in the equilibrium level of fraudulent data in the stolen data market. Effectively, a privacy regulation alleviates the lemons problem in the illicit market.

Empirically, I employ a unique dataset of stolen data packages posted in the market. Each observation is of an individual data package and contains information on the organization the data originated from, as well as the amount and types of data included. It is important to note that these data only cover what is available online, not necessarily everything that was stolen in a given data breach. The two may differ if a hacker decides to keep some data for themselves or that some of the data is not worth selling. Each package is labeled as being available before or after the GDPR, and whether the data it contains should have been protected by the GDPR. To the best of my knowledge, this is the first use of such data in the economics literature.

I combine the individual data packages to create a country-quarter level panel spanning January 2017 to November 2023 that tracks the number of data breaches and records available in stolen data markets that originate in a given country. I use this panel to estimate a difference-in-differences model measuring the effect of the GDPR on those two outcomes. I also break the post-GDPR period into short-run and long-run periods to measure if and how the effects changed over time. Short run is defined as one year after the regulation went into effect and long run is anytime after that.

I find that the GDPR caused the number of data breaches originating from regulated

countries to decrease by approximately 60 percent overall, with the long run decrease being slightly larger than the short run (61 percent versus 55 percent). Despite this, I find no statistically significant change in the number of records available. The granularity of my data allows me to estimate how the composition of the individual data packages changes to explain the lack of change in number of records.

At the individual data package level, I estimate how the content of the data packages—the number of records, amount of personally identifiable information (PII), and number of unique types of data—changes after the GDPR. As with the aggregate effects, I estimate both an overall change and separate short- and long-run changes. I find that the size of data packages, in terms of number of records, originating in regulated countries increases 68 percent in the long-run, while there is no statistically significant short-run change. The fraction of those records that are considered PII and number of unique data types in these packages do not change in any measured time period.

The increase in size of the data packages explains how the number of data packages could fall without an accompanying decrease in the number of records. This is in line with the key finding of the theoretical model. Additional empirical evidence of a shift towards larger targets is in the UK Cyber Security Breach Survey, which shows that small organizations (those with fewer than 50 employees) make up 95 percent of reported breaches in the 2017 survey, but only 48 percent in the 2022 survey. Large organizations (those with 250 or more employees) increase their share from less than one percent to approximately 24 percent (Department for Digital, Culture, Media and Sport, [2022](#)).

The effects of data privacy and security legislation have been studied in a number of contexts including healthcare (A. Miller and Tucker, [2009](#); A. Miller and Tucker, [2011](#); A. Miller and Tucker, [2018](#)) and online advertising (Goldfarb and Tucker, [2011](#)). The GDPR-specific literature covers its effects on firm performance (Koski and Valmari, [2020](#); C. Chen, Frey, and Presidente, [2022](#); Goldberg, Johnson, and Shriver, [2024](#)), competition (Johnson, Shriver, and Goldberg, [2023](#)), investment (J. Jia, Jin, and Wagman, [2021](#); Kircher and Foerderer,

2021; Janßen et al., 2022; Jia et al., 2025), and data collection (Aridor, Che, and Salz, 2021; Lukic, K. M. Miller, and Skiera, 2023; Demirer et al., 2024). These papers typically find the regulation had negative effects on business operations—decreases in competition, investment, and firm performance—or changes whose net welfare effects are more ambiguous, such as decreased data storage. Jones and Tonetti (2020) provide a theoretical backing to these results, showing that government policies restricting data use can increase privacy at the cost of reducing efficiency.

While I do not attempt to calculate the overall welfare effects of the policy, this paper does find a seemingly unambiguous benefit of the GDPR: the reduction in the number of data packages online. But even with this reduction, the extent to which individuals benefit is unclear given that there was not an accompanying reduction in the number of records available. It is possible that, while there are fewer breaches, those that remain contain enough information to leave the affected individuals no better off than before.

Significant work studying stolen data markets has been conducted by criminologists, who have derived some estimates of their sizes and products offered (Franklin et al., 2007; Holt and Lampke, 2010; Holt, Smirnova, and Chua, 2016). These papers conduct in-depth, descriptive studies of a handful of individual forums where data is sold. They do not study how public policy can have trickle-down effects on these markets. My unique dataset allows me to fill that gap in this area of the criminology literature, and extend the contribution into the economics of crime.

The model I present conceptually aligns with Becker (1968). Hackers decide whether to attempt a data breach based on the perceived costs and benefits of doing so. When the costs increase and the benefits decrease, there are fewer breaches. The costs and benefits are not perfectly observable, requiring hackers to base their actions on their beliefs of data collection practices and how well potential targets have protected themselves. This is similar to the mechanisms in Ayres and Levitt (1998) and Braakmann, Chevalier, and Wilson (2024). In Ayres and Levitt, car thieves could not observe which vehicles had tracking devices installed,

but were aware of which areas had higher installation rates. The higher likelihood of stealing a car that could be tracked caused them to steal fewer cars in those areas. In Braakmann, Chevalier, and Wilson, an increase in the price of gold motivates burglars to target homes in areas where homeowners were expected to store more gold. By causing a reduction in the expected value (akin to the Braakmann, Chevalier, and Wilson mechanism in reverse) and an increase in the expected cost (à la Ayres and Levitt) of breaching European organizations, the regulation incentivizes hackers to change who they target.

The remainder of this paper is structured as follows. Section II formally presents the model of the stolen data market discussed earlier. In Section III I describe the data used in this study. My empirical strategy is defined in Section IV and the results are presented in Section V. I offer concluding remarks on the implications and limitations of this study, and paths for future research in Section VI.

II. A Model of the Stolen Data Economy

Figure I Here

Figure I depicts how stolen data is produced and used. Legal data collectors—companies, schools, governments, and similar entities that hold customer, user, and employee data—initially decide what and how much information to collect. Collecting data comes at a cost. They must pay to gather it, keep it stored, and respond to user requests regarding their data. There is also the persistent risk of suffering a data breach and the additional costs that result: sending notifications to those affected, offering credit monitoring, performing security audits, legal costs, and fines imposed by governments. Organizations invest in cybersecurity to mitigate this risk. Prevention methods can be technical, consistently patching software vulnerabilities and encrypting data, or non-technical, teaching employees to detect phishing emails and improving their password management. The goal of these measures is to make it

more difficult for data to fall into the wrong hands.²

In the second stage, hackers choose which targets to breach. All else equal, targets who collect more (and more valuable) data are more attractive to hackers. Investing more in security makes targets less attractive.

Successfully stolen data is traded in online black markets found on platforms such as Telegram or data breach sites. Hackers, the market’s suppliers, advertise their products by describing what is in the data package and where it originated from (Holt and Lampke, 2010). Because they are anonymous, online, and illegal, these markets are especially vulnerable to adverse selection and asymmetric information problems. Many platforms have introduced reputation systems to solve such problems and facilitate trade.³ However, most transactions are still one-offs, limiting the effectiveness of these systems (Vu et al., 2020). A consequence of these information problems is that goods tend to sell for significantly less than their true value as buyers account for the risk of being scammed (Herley and Florêncio, 2010).

Privacy regulations are a negative supply shock in the market along two dimensions. First, they reduce the amount of data stored by organizations (Demirer et al., 2024), decreasing the benefit of breaching regulated targets. Second, they typically require increased investment in cybersecurity, increasing the cost of breaching regulated targets. Data from the United Kingdom Cyber Security Breaches Survey shows that nearly two-thirds of respondents made operational changes in response to the GDPR. Among those that made changes, 100 percent reported making changes related to the cybersecurity policies and practices (table I). Taken as a whole, privacy regulations increase the cost of acquiring the key input to the stolen data market: the data itself. This changes the incentives of hackers. The amount of data that can be extracted from a regulated organization decreases, while the marginal cost of breaching one increases, encouraging changes in the optimal effort allocation.

²This goal is not always achieved. A. Miller and Tucker (2011) find that use of encryption technology is actually associated with an increase in reports of data loss. They speculate this is due to firms and employees being less careful with the data they hold after it is encrypted.

³Some trading platforms have also created contract systems that set expectations for the parties involved in a transaction and help build suppliers’ reputation (Vu et al., 2020). Often, suppliers will give away their stolen data rather than sell it to help build their reputation.

Table I Here

To formalize the above, I propose a model of the stolen data economy with two parts: the stolen data market and legal data collectors. Buyers and sellers in the stolen data market trade data packages—bundles of data advertised as being stolen from legal data collectors. Legal data collectors choose how much data to collect and their security investment levels. The latter’s choices will be influenced by equilibrium in the stolen data market.

II.A The Stolen Data Market

Buyers in the stolen data market know that the data packages they purchase are either legitimate and worth V , or sold by a scammer and worth nothing. They have no way of distinguishing between legitimate and fake data prior to purchase. Based on their knowledge of the market, they form beliefs over the probability a data package is legitimate. Denoting the portion of legitimate data packages in the market q , the expected value of a bundle of data is

$$\mathbb{E}[V] = qV + (1 - q)0 = qV$$

Buyers have a skill parameter κ that determines how much benefit they receive from a given data package. For a bundle of data purchased at price p , the expected net utility received is

$$U_b = \kappa qV - p$$

For simplicity, assume $\kappa \sim \text{Uniform}[0, 1]$ and that buyers will only buy a bundle of data if the expected utility is non-negative, the market demand from a unit mass of buyers is

$$Q^D(p, q, V) = 1 - \frac{p}{qV} \tag{1}$$

Legitimate sellers are heterogeneous in their ability, γ , which determines their cost of

hacking an organization given its security investment (S). The profit earned by a legitimate seller on the sale of a data package is:

$$U_h = p - \gamma S$$

Again for simplicity, assume $\gamma \sim \text{Uniform}[0, 1]$. The portion of hackers that can earn profit from, and therefore want to breach, a target is $\frac{p}{S}$. With a unit mass of legitimate sellers, total legitimate supply will be

$$Q_\ell^S(p, S) = \frac{p}{S}$$

Scammers are included in the model via a single representative scammer that can produce fake data at a cost, which I normalize to zero.⁴ The scammer is aware that the more fake data they create and attempt to sell, the less buyers are willing to pay. Factoring in this effect, they will choose the optimal amount of fake data to create, effectively choosing q . The scammer's profit maximization problem is

$$\max_q (1 - q) \left(1 - \frac{p}{qV} \right) p$$

Solving this yields the optimal q given the market price and value of the data being faked:

$$q = \sqrt{\frac{p}{V}} \tag{2}$$

Defining Γ as the quantity of fake data bundles the scammer puts into the market, total market supply is

$$Q^S(p, S) = Q_\ell^S(p, S) + \Gamma$$

Equilibrium in the stolen data market consists of two conditions:

⁴I am assuming that the cost of generating fake data is fixed and does not depend on the value of data they are impersonating. Allowing the cost of fake data generation to vary with data quality would complicate the problem without adding to the intuition.

1. Buyer beliefs over q are accurate i.e.,

$$q = \frac{Q_\ell^S}{Q^S} = \frac{p/S}{p/S + \Gamma} \quad (3)$$

2. The market clears: $Q^S = Q^D$.

Under the second condition, the denominator of equation 3 can be replaced by equation 1 and the result rearranged to isolate q :

$$q = \frac{p}{V} + \frac{p}{S} \quad (4)$$

By the first equilibrium condition, the equilibrium price as a function of the target's security and value is found by setting equations 2 and 4 equal:

$$p^* = \frac{S^2 V}{(S + V)^2} \quad (5)$$

This price is subsequently used to find equilibrium q^* :

$$q^* = \frac{S}{S + V}$$

II.B Legal Data Collectors—Pre-GDPR

The stolen data market's equilibrium determines the probability a legal data collector is breached. Given their choice of data collection and security, the probability a legal data collector is breached is equal to the fraction of legitimate sellers that find it profitable to hack them. Plugging the market price given by equation 5 into the legitimate supply curve, breach probability, $\pi(S, V)$, is

$$\pi(S, V) = \frac{p^*}{S} = \frac{SV}{(S + V)^2} \quad (6)$$

If breached, legal data collectors will incur a loss of

$$L(S, V) = \frac{\Lambda V^\psi}{\delta S}, \quad 0 < \psi < 2$$

Considering the costs and benefits of data collection, the legal data collector optimization problem is

$$\max_{S, V} AV^\phi - \pi(S, V)L(S, V) - C(V) - C_S(S), \quad 0 < \phi < 1$$

where $C(V)$ and $C_S(S)$ are the cost of data collection and security, respectively, and A is a productivity term.

Assume that the value of data collection takes the functional form:

$$V = (\alpha_1 d_1^\rho + \cdots + \alpha_J d_J^\rho)^{1/\rho}$$

where d_i is the amount of data of type i collected. To simplify the optimization, define the security-to-data value ratio as $r = \frac{S}{V}$. Finally, define the two cost functions as:

$$C(V) = \sum_{j=1}^J \omega_j d_j, \quad C_S(S) = \omega_S S = \omega r V$$

Using these definitions, substituting in $S = rV$ for S , and reducing the problem to using only two data types for simplicity, the optimization problem becomes

$$\max_{r, d_1, d_2} A (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\phi/\rho} - \frac{\Lambda}{\delta (1+r)^2} (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\frac{\psi-1}{\rho}} - \sum_{j=1}^2 \omega_j d_j - \omega_S r (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{1/\rho}$$

Taking the first order conditions and simplifying yields the conditional factor demands for

d_1 , d_2 , and r :⁵

$$d_1(V) = \frac{V}{B_1^{1/\rho}}, \quad d_2(V) = \frac{V}{B_2^{1/\rho}},$$

$$r(V) = \left[\frac{2\Lambda}{\delta\omega_S} \right]^{\frac{1}{3}} V^{\frac{\psi-2}{3}} - 1$$

The legal data collector's optimal V is found by setting the marginal benefit of V equal to its marginal cost:⁶

$$\underbrace{\underbrace{\phi AV^{\phi-1}}_{MB} = \frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + \left(\omega_S \left[\frac{2\Lambda}{\delta\omega_S} \right]^{1/3} + \frac{\Lambda}{\delta \left[\frac{2\Lambda}{\delta\omega_S} \right]^{2/3}} \right) \frac{\psi+1}{3} V^{\frac{\psi-2}{3}} - \omega_S}_{MC}}_{MC} \quad (7)$$

This does not have a closed form solution, but one is not necessary to compute the relevant comparative statics in section II.D.

II.C Legal Data Collectors—Post-GDPR

For the purpose of this paper, the GDPR had two main effects: increasing the cost of data collection (ω_1 , ω_2 , and Λ) and adding a minimum security constraint that is a function of the value of data collected.⁷ Modeling the latter as a minimum security-to-data value ratio constraint, $\underline{r}$, the post-GDPR legal data collector optimization problem is:

⁵Derivations are in section A of the appendix. The B_i terms are:

$$B_1 = \left[\alpha_2 \left[\frac{\omega_2 \alpha_1}{\omega_1 \alpha_2} \right]^{\frac{\rho}{\rho-1}} + \alpha_1 \right], \quad B_2 = \left[\alpha_1 \left[\frac{\omega_1 \alpha_2}{\omega_2 \alpha_1} \right]^{\frac{\rho}{\rho-1}} + \alpha_2 \right]$$

⁶Derivations are once again in section A of the appendix

⁷The increasing cost of data collection is a product of the GDPR's additional obligations for data collectors, such as requiring impact assessments prior to data processing and sending data breach notifications. The minimum security requirement is motivated by Article 32, which requires organizations “implement appropriate technical and organisational measures to ensure a level of security appropriate to the risk” and lists four specific measures.

$$\max_{r, d_1, d_2} A(\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\phi/\rho} - \frac{\Lambda}{\delta(1+r)^2} (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\frac{\psi-1}{\rho}} - \sum_{j=1}^2 \omega_j d_j - \omega_S r (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{1/\rho}$$

s.t. $r \geq \underline{r}$

The conditional factor demands for d_1 and d_2 are unchanged, as is the conditional factor demand for r when the constraint does not bind. When the constraint does bind, $r = \underline{r}$, which changes the marginal cost of V . The constrained legal data collectors' new V^* solves

$$\underbrace{\phi AV^{\phi-1}}_{MB} = \underbrace{\frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + (\psi-1) \frac{\Lambda}{\delta(1+\underline{r})^2} V^{\psi-2} + \omega_S \underline{r}}_{MC}$$

This also only has a closed form solution under a specific value of ϕ , which I will again decline to assume.

II.D Comparative Statics: The GDPR Effect

There are three GDPR-driven changes of interest: the change in the number of breaches, the change in average breach size (in terms of the amount of data), and the change in the total amount of data available in the market. These will each be affected by both the increase in data collection costs and the security constraint.

Focusing first on the security constraint, two features of the legal data collector's optimization solution are important for determining the GDPR's effect: V^* is monotonically increasing in A , and r^* is monotonically decreasing in A .⁸

Substituting $S = rV$ into the breach probability function,

$$\pi = \frac{SV}{(S+V)^2} = \frac{r}{(1+r)^2}$$

⁸Derivations are again in section A of the appendix.

This is non-monotonic in r (see figure II) and, therefore, A . As r increases (and A decreases), π initially increases before reaching its maximum and falling.

Figure II Here

Defining $A = A^\dagger$ as the value of A where $r(A^\dagger) = \underline{r}$, all targets with $A < A^\dagger$ will be unaffected by the constraints.⁹ All others will need to increase their security ratio to $\underline{r}$, giving each constrained target a breach probability of $\pi(\underline{r}) \equiv \bar{\pi}$. A convenient property of the π function is that $\pi(r) = \pi(\frac{1}{r})$. Using this, define $A^{\dagger\dagger}$ as the value of A such that $\pi(r(A^{\dagger\dagger})) = \pi(\frac{1}{\underline{r}})$. If $\underline{r} > 1$, the security constraint will reduce breach probability for targets with $A \in (A^\dagger, A^{\dagger\dagger})$, but *increase* breach probability for those with $A > A^{\dagger\dagger}$ (see figure II).

The overall effect of the GDPR on the expected number of breaches therefore depends on A 's distribution, $F(A)$, and which of these changes dominates. If a sufficiently small mass of legal data collectors falls in the $A > A^{\dagger\dagger}$ group, the expected number of breaches will fall. Regardless of the exact distribution of A , this changes in the composition of breaches: higher A targets will make up a larger portion of the post-GDPR breaches than pre-GDPR breaches. These are organizations with high levels of data collection.

The expected amount of data in the market depends on both π and the total amount of data collected $D(A) = d_1(A) + d_2(A)$:

$$\mathbb{E}[Data] = \int \pi(A)D(A)dF(A)$$

While the effect of the security constraint on π is ambiguous, it is not for total data collection. Imposing higher security requirements increases the marginal cost of data collection, reducing optimal $D(A)$. Increasing the cost of data collection will similarly reduce $D(A)$.

The GDPR's total effect on the amount of data in the market depends on the relative sizes of the aforementioned compositional change and the reduction in data collection. The shift

⁹To the extent organization size correlates with amount of data held, this partitioning explains why only 61 percent of small organizations in the UK Cyber Security Breaches Survey reported making operational changes after the GDPR, compared to 90 and 95 percent of medium and large organizations (table I).

towards more data-rich targets increases the expected amount of data, while the decrease in data collection by all targets reduces the expected data quantity. The net change in total data on the market will depend on which of these effects dominates, but the latter has some nuance. Each type of target reduces its data collection, including the now more prominent high A types. But if their reduction in data collection is small enough, they will still have more data than the mid- A types had prior to the GDPR. If this is the case, the composition effect still increases the expected amount of data stolen. The same logic applies to determining how the expected breach size changes. If the composition effect dominates, the average amount of data in a breach increases.

To understand the driving force behind the composition effect, return to the stolen data market. Consider the price legitimate sellers can expect to receive from a given data package and how that price changes with r :

$$p^* = \frac{S^2V}{(S+V)^2} = \frac{r^2V}{(1+r)^2}$$

$$\frac{\partial p}{\partial r} = \frac{2rV(1+r) - 2r^2V(1+r)}{(1+r)^4} = \frac{2rV}{(1+r)^3}$$

For $r > 0$, the r values of interest, price is increasing in r . Price increases in r because q , the portion of legitimate sellers, is also increasing in r (figure III):

$$q = \frac{S}{S+V} = \frac{r}{1+r}$$

$$\frac{\partial q}{\partial r} = \frac{1}{(1+r)^2}$$

As r increases, the portion of sellers that are legitimate increases. This increases demand for the good, raising prices and incentivizing more breaches of data-rich targets. Effectively, the regulation alleviates the stolen data market's lemons problem enough to bring more large data breaches into the market.

Figure III Here

Summarizing the above, the model makes the following predictions about each outcome of interest.

Total number of breaches will decrease if the number of data-rich targets is small enough. The probability the most data-rich targets are breached increases after the policy because of the downstream effects in the stolen data market. In that market, privacy regulations reduce the equilibrium number of scammers, increasing the price of the data packages taken from these larger targets. However, breach probability falls or remains constant for all other targets.

The average breach size effect depends on two factors. First, the increase in breach probability for the most data-rich and decrease in breach probability for all other targets changes the composition of stolen data packages and increases expected breach size. But the reduction in data collection by all targets decreases expected breach size. The overall effect will depend on whether the composition or data reduction effect dominates.

The total amount of data effect will similarly depend on whether the composition or data reduction effect dominates. If the composition effect dominates and the most data-rich targets collect more data post-regulation than the targets whose post-regulation breach probability decreases did prior to the regulation, the total amount of data in the market will increase.

III. Stolen Data Market Observations

Data for this study come primarily from SpyCloud, a private cybersecurity company specializing in identity threat protection. They have constructed a catalog of data breaches gathered from online stolen data marketplaces. Each observation is of a data package posted in the market and contains information on where the data originated (either the country or specific organization), what types of data were stolen, and the total number of records included in the breach. The data packages were available online between 2015 and 2023,

though the breaches they originate from may have occurred as early as 2002. To the best of my knowledge, this is the first time such a dataset has been used to quantitatively study the effect of any policy change on the stolen data market. The details of the data allow me to go deeper than the aggregate and summary statistics previous research has depended on to see what is actually being traded. Unfortunately, I do not observe prices for all but a handful of data packages, restricting this paper to measuring only quantity effects.

Table II Here

Panel A of table II displays summary statistics for the three outcomes of interest in the study at the data package level: the total number of records in a data package; the fraction of the data in a data package that is personally identifiable information (PII); and the number of data types in each package. Examples of observed data types are email addresses, credit card information, or whether the identified person owns a cat. PII has multiple legal definitions, but is broadly any information that can identify an individual and may not be publicly known. A more in-depth discussion of the definition of PII is in section A of the online appendix.

Data packages vary greatly in terms of size, measured by the number of records. The largest contains over 700 million data points, while the smallest contains only one. Similarly, they range from having only one type of data to 55. Where they are more alike is in the fraction of records in the breach that are personally identifiable information. The 25th percentile breach is 50 percent PII, and the 75th percentile breach has 85 percent PII. Emails, considered PII under the GDPR, are the most common type of data in these breaches, closely followed by passwords (figure IV).

Figure IV Here

Because the GDPR applies to any entity collecting data on EU residents, not just those located in the EU, identifying treatment and control groups is difficult. For each data

package used in the analysis, I observe either the country from which the data originate or the name of the organization that was breached, and both for a subset of the observations. When I observe only the originating country, I assign the breach to that country. This makes treatment categorization simple: if the data originates in the EU, the package is treated. In the cases where I only observe the organization from which the data were stolen, I use one of two processes. First, I attempt to determine where the organization is headquartered. If they are an EU-based organization, the package is treated. If they are not, I search their website (where available) to see whether they indicate having a European presence. Those that do are categorized as treated. In addition to the manual search, I use a method similar to Demirer et al. (2024), who use firms’ data server locations to categorize them into treatment and control units, to determine country. I cannot observe data center locations, so instead I use the locations of the servers on which they host their websites, as that location is endogenous to the location of an organization’s users and customers.¹⁰

There are two pieces of the internet’s architecture key to connecting users with websites that allow me to observe where sites are physically located: DNS and GeoDNS. A Domain Name System (DNS) is essentially a phonebook for the internet. When a user types a domain name (e.g., www.fangraphs.com) into their web browser, it sends a query to the DNS, which then finds the IP address of the server hosting that website and connects it to the user. A GeoDNS does the same while taking into account the location of the user sending the initial query. For websites hosted in multiple locations, it will respond with the IP address of the server hosting the requested website that is closest to the user. As an example, suppose a website is hosted on one server in San Francisco, California, and another in Berlin, Germany.

¹⁰Although it has largely been abstracted away, the internet is fundamentally a physical network. Data flows through fiber optic cables that span across oceans and continents to deliver content to users. This means the further a user is physically located from the server hosting the content, the longer it takes content to be delivered. The difference in time may only be fractions of a second, but that can still have a noticeable effect on outcomes organizations care about. Previous research has found that a 0.1 second improvement in website load time can increase spending on retail sites by almost 10% (Deloitte, 2020). For streaming and gaming sites, decreasing lag time improves user experience and can be used as a differentiating factor. There is a clear incentive for organizations to host their website on servers that are physically near their users to minimize load time.

A user in Los Angeles will be connected to the San Francisco server, and a user in Frankfurt will be connected to the Berlin server.

To find where an organization hosts their website, I use the GeoNet API tool from Shodan, an internet devices research company.¹¹ The GeoNet API allows me to send GeoDNS queries from six locations around the world to any website and record the IP addresses that respond to each request.¹² I conduct these queries for each data package in my sample with an associated organizational website. After collecting the IP addresses of the responding servers, I use Shodan’s IP address lookup tool to find the physical location of each one. Under this method, I categorize a breach as having come from a regulated entity if the organization hosts their website on at least one server in the EU. An organization that hosts their website both in and outside the EU will also be considered regulated. Data packages that have not been manually assigned to a specific country are assigned to the country in which their originating organization hosts a majority of their servers. Panel B of table II breaks down the number of data packages that fall into each category before and after the GDPR.

Data packages are only included in the final dataset if I can determine whether the originating organization is subject to the GDPR (or was in the cases where the organization is no longer active), and when the data are available online (pre- or post-GDPR). This sample represents only data packages that are posted online, not necessarily all data that is collected or stolen. It is possible that some stolen data packages are not traded, in which case I cannot observe them.¹³ There are many reasons why a package may be unobservable. The hacker may decide they can profit more from using the data themselves than from publishing it. Or the hacker may have full access to the organization’s data, but decide only a subset is worth taking and selling. In the case of ransomware, the victim organization may decide to pay the ransom to prevent their data from being leaked.

Using the country assigned to each package and the date it was available online, I aggre-

¹¹<https://geonet.shodan.io>

¹²Requests are sent from servers in the United States, England, the Netherlands, Germany, India, and Singapore.

¹³If only part of a data package is traded, I only observe what is traded, not everything that was stolen.

gate the individual packages into a country-quarter level panel. The panel spans January 2017 to November 2023. Each country can be thought of as a market in the theoretical model in Section II. The value and cost of hacking organizations in regulated countries will be affected by the GDPR, and will remain unchanged in unregulated countries. Choosing January 2017 to November 2023 as the study’s time frame means that data packages available online prior to 2017 are not included in the panel, even though I can observe full information on them. I make the choice to exclude pre-2017 periods because SpyCloud was started in 2016. This padding removes any bias that may occur if the packages collected early in their operation are fundamentally different from the ones discovered later. For consistency in the sample I also exclude these observations from the primary data package level analysis. To control for population in the analysis, I add annual population data from the World Bank to the panel.

For robustness checks, I construct additional panels excluding any period after March 2020—to remove any bias introduced by the COVID-19 pandemic, and any data package originating from a multinational organization. The latter removes any bias that may arise due to partially treated organizations.

Not every country experiences a breach in every period. For those observations, I assign a value of zero to the two outcome variables: number of data breaches and number of records. As shown in figures V and VI, this creates mass points at zero for both variables. I discuss the implications of this for my estimation strategy in section IV.

Figure V Here

Figure VI Here

Roughly 27 percent of all country-quarters have a positive number of breaches (panel A of table III), and among the positive observations there are an average of six breaches and 21.6 million records stolen (panel B of table III). There is a large variation in both outcomes with as many as 245 breaches occurring and over one billion records being available in a quarter.

Figures VIIa and VIIb show how the number of data breaches and number of records trended over time. There is a clear decline in the number of data breaches immediately after the GDPR went into effect, but no obvious and persistent change in the number of records becoming available in each quarter. The fraction of total breaches originating in the EU also falls following the GDPR (figure VIII).

Table III Here

Figure VIII Here

IV. Empirical Strategy

I estimate the effects of the GDPR on aggregate quantities in the stolen data market, and the contents of the individual data packages traded. The model I propose in section II predicts that there will be a decrease in the total number of data packages originating in the EU. The change in the total amount of data available and average size of a data breach is ambiguous.

Treatment status in all cases is determined by where the data was originally collected and the date the data package was available online. A data package is in the treatment group if it originated in the EU or was stolen from an organization subject to the GDPR and became available in June 2018 or later. This definition includes multinational organizations, such as large social media organizations, as treated if they have any users in the EU. As discussed in Demirer et al. (2024), this complicates identification because these organizations may respond differently to the GDPR. The data they hold is partially treated since they likely hold information on individuals inside and outside the EU. Robustness checks excluding data packages from multinationals are in section D.B of the online appendix, and their findings are discussed in sections V.A and V.B.

IV.A Number of Data Packages and Quantity of Data Available

The two aggregate effects of interest are the change in the number of data packages and total amount of data available. They are estimated using the country-quarter panel described in section III. Each observation of country i is the aggregate of the individual data packages originating from that country in time period t . Most countries do not have a positive number of breaches in each period, creating a mass point at zero (figures V and VI). Privacy regulations are expected to affect the extensive margin because they change the relative value of breaching organizations in regulated countries, making them less likely to have a positive number of breaches in a given period. To measure the extensive margin effect, I estimate the linear probability model:

$$Positive_{it} = \gamma_i + \tau_t + \delta D_i \times Post-GDPR_t + \varepsilon_{it}.$$

where γ_i and τ_t are country and quarter fixed effects. D_i equals one if the country is in the EU, and $Post-GDPR_t$ equals one if the period is after the second quarter of 2018. The dependent variable, $Positive_{it}$, is an indicator for whether country i has at least one breach in period t .

To measure the impact of the GDPR on the number of breaches and total number of records available, I use a Poisson model:

$$Y_{it} = \exp(\gamma_i + \tau_t + \delta^{Agg} D_i \times Post-GDPR_t - \log(population_{it})) \varepsilon_{it} \quad (8)$$

where γ_i , τ_t , D_i , and $Post-GDPR_t$ are all defined as in the extensive model. To explicitly obtain the percentage change in the outcome, δ^{Agg} must be transformed:

$$\delta^{Agg\%} = \exp(\delta^{Agg}) - 1. \quad (9)$$

Standard errors are clustered at the country level. The offset, $\log(population)$, is used to

account for difference in sizes among the countries.

I use a Poisson model rather than a log-like transformation because of the mass points at zero. In order to use log-like transformations on the outcomes, it would be necessary to either add a constant to each observation or use a transformation that is defined at zero, such as the inverse hyperbolic sine, to include the full sample in the estimation. Mullahy and Norton (2024) show that log-like transformations significantly change the estimated marginal effects when zero mass points are present. Further, J. Chen and Roth (2023) find that, in the presence of zero mass points, if the treatment has extensive margin effects, the estimated average treatment effect is sensitive to the units of the outcome variable, making the interpretation of the estimates difficult. The model I present in section II makes clear that privacy regulations should affect the extensive margin as it changes the probability organizations in regulated countries are breached, changing the likelihood those countries experience a positive number of breaches.

To test whether the effect changes over time, I break the *Post-GDPR_t* term into short- and long-run effects, estimating:

$$Y_{it} = \exp \left(\gamma_i + \tau_t + \delta_{SR}^{Agg} Short-Run_t \times D_i + \delta_{LR}^{Agg} Long-Run_t \times D_i - \log(population_{it}) \right) \varepsilon_{it} \quad (10)$$

where *Short-Run_t* equals one when $t \in \{\text{June 2018} - \text{May 2019}\}$ and *Long-Run_t* equals one for all periods after May 2019.¹⁴ This allows for changes in the behavior of both those collecting and stealing data that lags behind the policy implementation. Organizations may increase their compliance level, and hackers may reevaluate their strategies after observing organizational behavior changes.

The identifying assumptions underlying these models are conditional no anticipation, and that the growth rate between periods the treated group would have realized in the absence

¹⁴The short- and long-run definitions follow Demirer et al. (2024).

of treatment is the same as that experienced by the control group, i.e., there are parallel trends in the ratio of outcomes between periods (Wooldridge, 2023):

$$\frac{E[Y^{Post}(0)|D = 1]}{E[Y^{Pre}(0)|D = 1]} = \frac{E[Y^{Post}(0)|D = 0]}{E[Y^{Pre}(0)|D = 0]}.$$

To test this assumption, I estimate an event study model:

$$Y_{it} = \exp \left(\gamma_i + \tau_t + \sum_{t \neq -1} \delta_{it}^{Agg} D_i \times Post-GDPR_t - \log(population) \right) \varepsilon_{it} \quad (11)$$

where all notation is defined as before and standard errors are once again clustered at the country level.

Under the model in section II, the increase in cost and decrease in value of breaching regulated organizations caused by the GDPR should cause the number of data breaches originating in regulated countries to decrease. All else equal, the number of records should decrease as well, but changes in which targets are hacked and which data packages are subsequently sold may blunt this effect. If there is a sufficiently large shift towards data-rich targets post-GDPR, the expected number of records may even increase.

IV.B Data Package Content Effects

Effects on individual data packages are estimated using the linear model:

$$y_j = \gamma_j + \tau_t + \delta^{DP} D_j \times Post-GDPR_t + \varepsilon_j \quad (12)$$

where D_j equals one if the package originated from a regulated organization, and $Post-GDPR_t$ indicates whether the data package was available June 1, 2018 or later. I use June 1, 2018 as the beginning treatment date, rather than the day the GDPR was enforced, to allow for a lag between when data became available online and when it was stolen.

The three outcomes of interest are the log of the total number of records in the package,

the amount of personally identifiable information (PII), and the number of unique types of data in the package. The parameter of interest is δ^{DP} .

I once again break the *Post-GDPR* term into short- and long-run effects and estimate:

$$y_j = \gamma_j + \tau_t + \delta_{SR}^{DP} D_j \times Short-Run_t + \delta_{LR}^{DP} D_j \times Long-Run_t + \varepsilon_j \quad (13)$$

where *Short-Run_t* and *Long-Run_t* are defined as they were in the aggregate effects section.

The expected effects on individual data packages are ambiguous under the model in section II. If the GDPR increases the portion of breaches originating from data-rich organizations enough, then the expected amount of data in a data breach increases. Given the restrictions on collecting PII, the fraction of all records that are PII might be expected to decrease, but that will depend on the effects of the regulation on non-PII data collection as well.

V. Results

The main results are presented in sections V.A and V.B. Robustness checks and alternative model specifications are discussed in section V.C.

V.A Number of Data Packages and Quantity of Data Available

On the extensive margin, I find the GDPR is associated with a roughly 20 percent decline in the probability of finding a data breach that originates from a regulated country online (table IV). This effect is larger in the long-run than in the short-run (-21.1 percent versus -15.8 percent, respectively).

Table IV Here

The aggregate treatment effects on the number of breaches and total amount of data posted from a country are presented in table V. I find that the number of data breaches

fell 54.8 percent and 60.9 percent in the short and long run, respectively, and 60.4 percent overall. This result is consistent with the predicted effects of both a decrease in the amount of data collected and an increase in security investment by regulated organizations on the market. Fewer organizations are worth hacking, so there is a decrease in the number of data breaches. The same logic applies to my extensive margin findings.

Table V Here

Despite the large decrease in the number of breaches, I find no statistically significant change in the number of records in the market. Interestingly, the short- and long-run effect estimates indicate there may have been an initial decrease in the aggregate number of records available, followed by an increase in the long run. Intuitively, this could be the result of hackers making adjustments to their strategies in the long run. A shift in strategy towards higher-valued targets would result in larger data packages becoming a larger share of observed postings, which is mechanically necessary for these results to be true.

Event study plots to provide evidence that the parallel trends assumption holds are in figure IX. For number of data breaches, the coefficient estimates for each period prior to the GDPR have zero in the 95 percent confidence interval, while post-GDPR there is a clear decrease in the number of data breaches (figure IXa). Each period of the event study shows no statistically significant effect on the number of records traded (figure IXb).

Figure IX here

V.B Individual Data Package Content Effects

Figure X Here

At the individual breach level, I find that data packages originating in regulated organizations increased in size by 62.9 percent, as measured by number of records they contain (column three of table VI). This effect is driven by long-run changes, with there being no statistically significant short-run effect (column four of table VI).

An increase in the size of the data packages is counterintuitive on its face. If data privacy legislation successfully reduces data collection, which it appears to do, then it seems natural that there would be a corresponding reduction in the number of records included in the packages. Less data collected means there is less data to steal. However, if the GDPR brings more high-value breaches into the market, then the expected value of the remaining breaches increases even after accounting for the change in value caused by the GDPR. These breaches would presumably contain larger amounts of data, increasing the expected number of records in any given breach. Figure X shows that the distribution of the number of records in a breach shifted right after the GDPR for both the treated and untreated groups. The main difference is in the tail end of distributions. The untreated group appears to have relatively more weight at the lower end of the distribution, while the treated group has more weight at the very high end.

Looking specifically at the amount of PII in a breach, I find that the number of records that constitute PII increased by 60.6 percent in the long run (table VII). Given that most of the data in the packages qualifies as PII (table II), this is expected with the increase in the overall number of records per package.

These are the only statistically significant changes at the data package level. I find no change in the fraction of records that are PII (table VIII) or number of unique types of data in the packages (table IX). One potential explanation for this is that only certain types of data have value in the stolen data market. If the data no longer collected by regulated organizations is not considered valuable in this other market, it is unlikely that there would be an effect on data package contents beyond their size.

Table VI Here

Table VII Here

Table VIII Here

Table IX Here

V.C Robustness

To check the robustness of my results, I re-estimated the aggregate effects using a number of alternative samples and model specifications. Splitting the sample into small and large countries, defined by whether the population is above or below the median in 2018, I find that the point estimate is slightly more negative for small countries than for large countries, but the directional effects are the same. The only notable change is that the short-run effect in large countries is statistically insignificant.¹⁵

The effects on the number of data packages and total quantity of data available also hold under different model and panel construction choices. None of removing the population offset, weighting the observations by population, or changing the outcome variables to per capita variables changes the direction or significance of the results.¹⁶ Removing all observations from Brazil and China, which each implemented privacy regulations of their own near the end of the study, has no effect. Nor does removing all multinational breaches from the aggregate numbers or the post-COVID-19 periods.

Using log-like transformations, $\text{Log}(Y + 1)$ and the inverse hyperbolic sine, as alternatives to the Poisson model changes the magnitude of the effect on both aggregate outcomes and shows there is a significant decrease in the number of records available. However, these outcome changes are all consistent with the bias J. Chen and Roth, 2023 and Mullahy and Norton, 2024 show log-like transformations introduce when used on data with a mass point at zero.

The data package content effects are also robust to changes in how the sample is constructed.

A more in-depth discussion of these robustness checks can be found in the online appendix.

¹⁵I define a country as small or large based on whether their population is below or above the median country population in 2018.

¹⁶It should be noted that converting the outcomes to per capita terms changes them from discrete to continuous variables, making a Poisson model inappropriate to use.

VI. Discussion and Conclusion

As organizations continue to collect large amounts of data, the risk of that data being stolen and sold will be ever-present. In this paper I have shown that data protection regulations can have a significant effect on the illicit market for data.

I estimate that the GDPR is associated with a 60 percent reduction in the number of data breaches originating in EU countries available in stolen data markets. There is, however, no accompanying reduction in the number of individual records in these markets, as the size of data packages increased over 60 percent as well. I find no other changes in the contents of the data packages.

These results are consistent with what the model in section II predicts will happen after a privacy regulation goes into effect. Although a privacy regulation reduces breach probability for some legal data collectors, it will increase breach probability for the extremely data-rich legal data collectors. This is because the regulation alleviates the lemons problem in the stolen data market, incentivizing more hacks against larger targets. The result is that even though the expected number of breaches falls, the expected total amount of data and average breach size may increase.

This paper partially fills the gap in the literature on privacy regulation, and the GDPR in particular, regarding potential benefits of these regulations. It is the first to study the effects of privacy regulation on the stolen data market and show a causal impact.

There are many paths forward for future research on this topic. Qualitative and quantitative work on the abilities and behaviors of hackers could provide insights into whether the effects I observe empirically are driven strictly by the changes in hacker incentives I propose, or if there are other factors, such as changes in hacker skill, at play.

This paper is missing a key component of the market: prices. Although there are many hurdles to collecting quality price data in these markets, doing so would open the door to a more complete analysis of their inner workings and the value hackers place on certain types

of information.

Implicit in my model is the assumption that hacker skill remains constant. Human capital accumulation on the part of hackers may be an equilibrium response to the GDPR. If hackers were to become more productive, the cost of hacking would decrease, resulting in more breaches, but the expected value may decrease as relatively low-value targets become viable marks now that they are cheaper to hack. That I find a decrease in the number of breaches and an increase in their quality (in terms of size) suggests this is not a concern. However, I do not observe any direct measure of hacker ability and therefore cannot fully rule out the possibility that it has changed.

Finally, while there have been a number of studies on the effects of the GDPR on specific firm outcomes, changes in data collection, and now cybercrime, there is still no overarching study of its overall welfare effects or how individual components of the policy influence outcomes of interest. Reducing the number of data breaches is likely beneficial to all not looking to buy or sell stolen data. But the extent of that benefit may be limited given that the number of records in stolen data markets did not also fall. With at least the same amount of data available if not more, individuals may be no better off than they were before. To test this, one would need to calculate how the probability of a person’s data being online has changed, or at least count the number of unique individuals with data in each breach, which I am unable to do with my data.

On the cyber criminal side of the problem, the GDPR may have made them better off in some cases. By incentivizing a shift in effort towards higher-value targets and potentially alleviating the stolen data market’s asymmetric information problem, the GDPR may have provided some benefit to internet miscreants.

A thorough welfare analysis will also need to account for the externalities of data collection (Choi, Jeon, and Kim, 2019; Acemoglu et al., 2022), and the larger costs of cyberattacks, such as ransomware attacks on hospitals (Neprash, McGlave, and Nikpay, 2026).

For policymakers, a key question is whether subsequent privacy regulations will have

effects similar to the GDPR, or if the reduction in breaches against the EU is really the result of cybercrime being displaced towards non-regulated targets. A spillover could be expected if hackers are effort-constrained and rank all potential targets, breaching only the top choices until they exhaust their effort endowment. Under this model, unregulated organizations become relatively more valuable targets, increasing the probability they are breached. The reduction in data breaches I observe would be a combination of fewer breaches against regulated organizations, and an increase in breaches against non-regulated ones. While I am not able to directly test for spillovers, an increase in the number of breaches against non-regulated targets would appear in the time series plots around the GDPR's passage. As can be seen in figure [VIIa](#), no such increase is apparent. This suggests there is no substantial displacement effect. With more countries considering and adopting data privacy regulations, research on this subject would have high returns in the debate over how to design future policy.

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Appendix

A. Model Derivations

A.A The Stolen Data Market

A.A.1 Demand

Demand in the stolen data market equals the probability a buyer finds a data package worth purchasing at a given price:

$$\Pr(\kappa\mu > p)$$

With the simplifying assumption that $\kappa \sim \text{Uniform}[0, 1]$, and the behavioral assumption that buyers will only purchase a data package if the expected utility is positive:

$$Q^D(p, \mu) = \Pr(\kappa\mu > p) = \Pr(\kappa > \frac{p}{\mu}) = 1 - \frac{p}{\mu}$$

A.A.2 Legitimate Supply

Legitimate supply in the market is provided by heterogeneous hackers with utility

$$U_h = p - \gamma S$$

As with demand, use the simplifying assumption that $\gamma \sim \text{Uniform}[0, 1]$ and the behavioral assumption that sellers only offer packages they expect to at least break even on, with a unit mass of legitimate hackers, total legitimate supply is:

$$Q_\ell^S(p, S) = \frac{p}{S}$$

A.A.3 Scammer Optimization

The profit maximizing representative scammer solves:

$$\max_q (1 - q) \left(1 - \frac{p}{qV}\right) p$$

Taking the first order condition and solving for q ,

$$\begin{aligned} -1 \left(1 - \frac{p}{qV}\right) p + (1 - q) p \left(\frac{p}{q^2 V}\right) &= 0 \\ p \left(1 - \frac{p}{qV}\right) &= \frac{p^2}{q^2 V} - \frac{p^2}{qV} \\ p &= \frac{p^2}{q^2 V} \\ q^* &= \sqrt{\frac{p}{V}} \end{aligned}$$

The equilibrium price and q are derived as in the main body of the paper:

$$p^* = \frac{S^2 V}{(S + V)^2} \quad q^* = \frac{S}{S + V}$$

A.B Legal Data Collectors

Legal data collector's breach probability is equal to the probability a legitimate hacker finds it profitable to target them. Using the supply curve and equilibrium price, this is:

$$\pi(S, V) = \frac{SV}{(S + V)^2}$$

Their loss function is

$$L(S, V) = \frac{\Lambda V^\psi}{\delta S}, \quad 0 < \psi < 2$$

Substituting $S = rV$ and simplifying yields a simple function for expected losses:

$$\mathbb{E}[Loss] = \pi(r, V)L(r, V) = \frac{rV^2}{(rV + V)^2} \frac{\Lambda V^\psi}{\delta r V} = \frac{\Lambda}{\delta(1+r)^2} V^{\psi-1}$$

With two types of data, the legal data collector optimization problem is

$$\max_{r, d_1, d_2} A(\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\phi/\rho} - \frac{\Lambda}{\delta(1+r)^2} (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\frac{\psi-1}{\rho}} - \sum_{j=1}^2 \omega_j d_j - \omega_S r (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{1/\rho}$$

Taking the first order conditions with respect to the d terms:

$$\begin{aligned} d_1 : \left[\frac{\phi}{\rho} A V^{\phi-\rho} - \frac{\Lambda}{\delta(1+r)^2} \frac{\psi-1}{\rho} V^{\psi-1-\rho} - \omega_S r \frac{1}{\rho} V^{1-\rho} \right] \alpha_1 \rho d_1^{\rho-1} &= \omega_1 \\ d_2 : \left[\frac{\phi}{\rho} A V^{\phi-\rho} - \frac{\Lambda}{\delta(1+r)^2} \frac{\psi-1}{\rho} V^{\psi-1-\rho} - \omega_S r \frac{1}{\rho} V^{1-\rho} \right] \alpha_2 \rho d_2^{\rho-1} &= \omega_2 \end{aligned}$$

Taking the ratio of the two,

$$\begin{aligned} \frac{\alpha_1 d_1^{\rho-1}}{\alpha_2 d_2^{\rho-1}} &= \frac{\omega_1}{\omega_2} \\ \rightarrow d_1 &= \left[\frac{\omega_1}{\omega_2} \frac{\alpha_2}{\alpha_1} \right]^{\frac{1}{\rho-1}} d_2 \end{aligned}$$

Plugging this into the value production function yields the conditional factor demand for d_1 :

$$\begin{aligned} V &= (\alpha_1 d_1^{*\rho} + \alpha_2 d_2^{*\rho})^{1/\rho} \\ V^\rho &= \underbrace{\left[\alpha_1 \left[\frac{\omega_1}{\omega_2} \frac{\alpha_2}{\alpha_1} \right]^{\frac{\rho}{\rho-1}} + \alpha_2 \right]}_{\equiv B_2} d_2^\rho \\ \rightarrow d_2(V) &= \frac{V}{B_2^{1/\rho}} \\ d_1(V) &= \left[\frac{\omega_1}{\omega_2} \frac{\alpha_2}{\alpha_1} \right]^{\frac{1}{\rho-1}} \frac{V}{B_2^{1/\rho}} \end{aligned}$$

By symmetry,

$$d_1(V) = \frac{V}{B_1^{1/\rho}}$$

where

$$B_1 = \left[\alpha_2 \left[\frac{\omega_2 \alpha_1}{\omega_1 \alpha_2} \right]^{\frac{\rho}{\rho-1}} + \alpha_1 \right]$$

Taking the first order condition with respect to r will then give r as a function of V :

$$\begin{aligned} \frac{2\Lambda}{\delta(1+r)^3} V^{\psi-1} &= \omega_S V \\ \left[\frac{2\Lambda}{\delta\omega_S} \right] V^{\psi-2} &= (1+r)^3 \\ r(V) &= \left[\frac{2\Lambda}{\delta\omega_S} \right]^{\frac{1}{3}} V^{\frac{\psi-2}{3}} - 1 \end{aligned}$$

To find the optimal V , consider the total cost of data collection:

$$\begin{aligned} TC(V) &= \omega_1 d_1(V) + \omega_2 d_2(V) + \omega_S r(V)V + \frac{\Lambda}{\delta(1+r(V))^2} V^{\psi-1} \\ &= \omega_1 d_1(V) + \omega_2 d_2(V) + \omega_S \left[\frac{2\Lambda}{\delta\omega_S} \right]^{\frac{1}{3}} V^{\frac{\psi+1}{3}} - \omega_S V + \frac{\Lambda}{\delta \left[\frac{2\Lambda}{\delta\omega_S} \right]^{2/3}} V^{\frac{\psi+1}{3}} \end{aligned}$$

The marginal cost of V is then

$$MC(V) = \frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + \left(\omega_S \left[\frac{2\Lambda}{\delta\omega_S} \right]^{1/3} + \frac{\Lambda}{\delta \left[\frac{2\Lambda}{\delta\omega_S} \right]^{2/3}} \right) \frac{\psi+1}{3} V^{\frac{\psi-2}{3}} - \omega_S \quad (14)$$

The marginal benefit of an additional unit of V is

$$\frac{\partial}{\partial V} AV^\phi = MB(V) = \phi AV^{\phi-1} \quad (15)$$

The optimal V is found by setting the marginal cost (equation 14) equal to the marginal benefit (equation 15).

$$\phi AV^{\phi-1} = \frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + \left(\omega_S \left[\frac{2\Lambda}{\delta\omega_S} \right]^{1/3} + \frac{\Lambda}{\delta \left[\frac{2\Lambda}{\delta\omega_S} \right]^{2/3}} \right) \frac{\psi+1}{3} V^{\frac{\psi-2}{3}} - \omega_S \quad (16)$$

This only has a closed form solution if $\phi - 1 = \frac{\psi - 2}{3}$, which I will refrain from assuming because it is not necessary to derive the needed insights from the model.

After the GDPR, the firm's optimization problem includes a constraint term:

$$\begin{aligned} \max_{r, d_1, d_2} A(\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\phi/\rho} - \frac{\Lambda}{\delta(1+r)^2} (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{\frac{\psi-1}{\rho}} - \sum_{j=1}^2 \omega_j d_j - \omega_S r (\alpha_1 d_1^\rho + \alpha_2 d_2^\rho)^{1/\rho} \\ + \lambda [r - \underline{r}] \end{aligned}$$

Taking the ratio of the first order conditions with respect to d_1 and d_2 results in the same equation and conditional factor demands as the unconstrained problem. Taking the first order condition with respect to r ,

$$\frac{2\Lambda}{\delta(1+r)^3} V^{\psi-1} - \omega_S V + \lambda = 0$$

When the constraint is not binding, $\lambda = 0$ and the solution is the same as the unconstrained solution. When the constraint does bind, $r = \underline{r}$. This changes the marginal benefit and marginal cost comparison. For constrained targets, the total cost of V is

$$TC(V) = \omega_1 d_1(V) + \omega_2 d_2(V) + \frac{\Lambda}{\delta(1+\underline{r})^2} V^{\psi-1} + \omega_S \underline{r} V$$

And the marginal cost

$$MC(V) = \frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + (\psi - 1) \frac{\Lambda}{\delta(1 + \underline{r})^2} V^{\psi-2} + \omega_S \underline{r}$$

Setting this equal to the marginal benefit,

$$\phi A V^{\phi-1} = \frac{\omega_1}{B_1^{1/\rho}} + \frac{\omega_2}{B_2^{1/\rho}} + (\psi - 1) \frac{\Lambda}{\delta(1 + \underline{r})^2} V^{\psi-2} + \omega_S \underline{r}$$

A closed form solution again only exists under certain conditions, and is still unnecessary to gain insights from the model.

A.B.1 GDPR Effect

There are three outcome changes driven by the GDPR we are interested in: the change in the expected number of breaches, the change in the expected amount of data available in the market, and the change in the average breach size. These will be affected by both the increase in data collection costs and the security constraint, but for the sake of simplicity I will address each separately.

Corollary A.0.1. *Assuming each target has an interior maximum and the distribution of A has a sufficiently high upper bound, there exists a unique $A = A^\dagger$ above which the security ratio constraint binds.*

Proof. Let

$$F(V, A) = MB(V) - MC(V) = 0$$

by the implicit function theorem:

$$\begin{aligned}\frac{\partial V}{\partial A} &= -\frac{\frac{\partial F}{\partial A}}{\frac{\partial F}{\partial V}} \\ \frac{\partial F}{\partial A} &= \phi V^{\phi-1} \\ \frac{\partial F}{\partial V} &= \phi(\phi-1)AV^{\phi-2} - \frac{\partial MC}{\partial V}\end{aligned}$$

$\frac{\partial F}{\partial A}$ is obviously positive, and $\frac{\partial F}{\partial V}$ is the second order condition of the target's objective function. If there is an interior maximum, this will be negative. Therefore, $\frac{\partial V}{\partial A} > 0$.

With that,

$$\frac{\partial r}{\partial A} = \left[\frac{2\Lambda}{\delta\omega_S} \right]^{1/3} \frac{\psi-2}{3} V^{\frac{\psi-5}{3}} \frac{\partial V}{\partial A}$$

Having assumed that $\psi < 2$, r is monotonically decreasing in A . □

Whether π is increasing or decreasing in r depends on r . If $r < 1$, π is increasing in r , while if $r > 1$, π is decreasing. Since r is monotonically decreasing in A , $r < 1$ will be true for targets with sufficiently high A . A property of π is that $\pi(r) = \pi(\frac{1}{r})$:

$$\begin{aligned}\pi(r) &= \frac{r}{(1+r)^2} \\ \pi\left(\frac{1}{r}\right) &= \frac{1/r}{(1+1/r)^2} \\ &= \frac{1/r}{\left(\frac{r+1}{r}\right)^2} \\ &= \frac{r}{(1+r)^2} = \pi(r)\end{aligned}$$

Defining $A = A^{\dagger\dagger}$ as the point where $\pi(r(A^{\dagger\dagger})) = \pi\left(\frac{1}{r}\right)$, if there is a sufficiently small mass of firms where $A > A^{\dagger\dagger}$, then the security requirement constraint will decrease the expected number of breaches.

Any decrease in the expected number of breaches is driven by a decrease in breaches of targets whose A falls between $A^\dagger$ and $A^{\dagger\dagger}$. This changes the composition of the data breaches. High A targets will take up a higher portion of breaches than prior to the GDPR. This compositional change will also impact the expected amount of data stolen.

The expected amount of breached data appearing in the market depends on both π and $D(A)$:

$$\mathbb{E}[Data] = \int \pi(A)D(A)dF(A)$$

I have already shown that whether the security ratio constraint causes $\pi(A)$ to increase or decrease will depend on whether A is greater than $A^{\dagger\dagger}$. Total data collection, D , is monotonically increasing in V , and therefore A . Requiring a higher r increases the marginal cost of data collection, reducing optimal data collection.

Whether the security constraint increases or decreases the amount of data available will depend on the relative sizes of the aforementioned compositional change and the reduction in data collection. The shift towards more data-rich targets increases the expected amount of data, while the decrease in data collection reduces expected data collection. The net change will depend on which of these effects dominates, but the latter has some nuance. Each type of target reduces its data collection, including the now more prominent high A types. But if their reduction in data collection is small enough, they may still have more data than the mid- A types had prior to the GDPR. If this is the case, the composition effect will still increase the expected amount of data stolen.

This same logic applies to determining how the expected breach size changes. The composition effect can still lead to larger breaches on average.

In the stolen data market, consider the price legitimate sellers can expect to receive from

a given data package and how that price changes with r :

$$p^* = \frac{S^2V}{(S+V)^2} = \frac{r^2V}{(1+r)^2}$$

$$\frac{\partial p}{\partial r} = \frac{2rV(1+r) - 2r^2V(1+r)}{(1+r)^4} = \frac{2rV}{(1+r)^3}$$

For $r > 0$, price is increasing in r . Price increases in r because q , the portion of legitimate sellers, is also increasing in r :

$$q = \frac{S}{S+V} = \frac{r}{1+r}$$

$$\frac{\partial q}{\partial r} = \frac{1}{(1+r)^2}$$

As r increases, the portion of sellers that are legitimate increases. This increases demand for the good, raising prices. Effectively, the GDPR alleviates the market's lemons problem for large data packages enough to bring more into the market.

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Tables

Table I: Percentage of Organizations Reporting Operational Changes Due to the GDPR

Survey Year	Any Change		Change in Cybersecurity	
	2018	2019	2018	2019
Overall	12.75%	63.71%	100.00%	100.00%
Small	9.91%	61.56%	100.00%	100.00%
Medium	28.71%	90.66%	100.00%	100.00%
Large	52.91%	95.81%	100.00%	100.00%

Note: Department for Digital, Culture, Media and Sport (2022), author’s calculations. Respondents were asked “*Has your organisation made any changes or not to the way you operate in response to GDPR?*” and “*Have any of these changes been related to your cyber security policies or processes, or not?*” The fraction of respondents answering yes to the first question is in the first two columns. The fraction answering yes to the second question, among those answering yes to the first, is in the last two columns.

Table II: Data Package Summary Statistics

Panel A: Data Package Summary Statistics

	Number of Records	PII Fraction	Number of Data Types
Observations	4,394	4,394	4,394
Mean	3,544,186	0.693	6.220
Std. Dev.	28,996,342	0.191	5.208
Min.	1	0	1
25%	5,164	0.500	2
50%	46,748	0.667	4
75%	288,555	0.858	9
Max.	716,409,393	1	55

Panel B: Group Counts

	Pre-GDPR N=1,621		Post-GDPR N=2,773	
	Non-EU	EU	Non-EU	EU
N	1,175	446	2,292	481

Table III: Panel Data Summary Statistics

Panel A: All Periods

	Number of Breaches	Number of Records (M)	> 0 Breaches
Observations	2,716	2,716	2,716
Mean	1.62	5.73	0.27
Std. Dev.	9.52	49.73	0.44
Min.	0	0.00	0
25%	0	0.00	0
50%	0	0.00	0
75%	1	0.00	1
Max.	245	1,009.74	1

Panel B: Non-Zero Periods Only

	Number of Breaches	Number of Records (M)
Observations	721	721
Mean	6.09	21.60
Std. Dev.	17.74	94.78
Min.	1	0.00
25%	1	0.04
50%	2	0.26
75%	4	1.90
Max.	245	1,009.74

Table IV: Extensive Margin Effects

	Dependent Variable: Positive Number of Breaches	
	(1)	(2)
Post $\times$ Treatment	-0.201*** (0.039)	
SR $\times$ Treatment		-0.158*** (0.049)
LR $\times$ Treatment		-0.211*** (0.039)
Observations	2,716	2,716
R^2	0.469	0.469
Period Fixed Effects	Y	Y
Country Fixed Effects	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level.

Table V: Aggregate Effects

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.927*** (0.266)		0.342 (0.429)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.794*** (0.297)		-0.221 (0.590)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.939*** (0.284)		0.407 (0.429)
$\hat{\delta}^{Agg\%}$	-0.604 (0.105)		0.407 (0.604)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.548 (0.134)		-0.198 (0.473)
$\hat{\delta}_{LR}^{Agg\%}$		-0.609 (0.111)		0.502 (0.645)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	0.792	0.793	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table VI: Data Package Effects: Number of Records

	Dependent Variable: Log(Number of Records)			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.936** (0.412)		0.488* (0.251)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.691 (0.491)		0.196 (0.333)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.963** (0.414)		0.519** (0.258)
Multinational			1.390*** (0.252)	1.392*** (0.252)
$\hat{\delta}^{DP\%}$	1.550 (1.051)		0.629 (0.408)	
$\hat{\delta}_{SR}^{DP\%}$		0.996 (0.981)		0.216 (0.405)
$\hat{\delta}_{LR}^{DP\%}$		1.618 (1.085)		0.680 (0.433)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.268	0.268	0.276	0.276

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. $\hat{\delta}^{DP\%}$ terms are calculated by transforming the $\hat{\delta}^{DP}$ terms. $\hat{\delta}^{DP\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{DP\%}) = \exp(\hat{\delta}^{DP})SE(\hat{\delta}^{DP})$.

Table VII: Data Package Effects: Number of PII Records

	Dependent Variable: Number of PII Records			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.933** (0.422)		0.441* (0.247)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.677 (0.542)		0.134 (0.384)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.960** (0.422)		0.474* (0.251)
Multinational			1.525*** (0.268)	1.527*** (0.269)
$\hat{\delta}^{DP\%}$	1.542 (1.073)		0.555 (0.384)	
$\hat{\delta}_{SR}^{DP\%}$		0.969 (1.068)		0.143 (0.439)
$\hat{\delta}_{LR}^{DP\%}$		1.613 (1.103)		0.606 (0.404)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.270	0.270	0.279	0.279

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. $\hat{\delta}^{DP\%}$ terms are calculated by transforming the $\hat{\delta}^{DP}$ terms. $\hat{\delta}^{DP\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{DP\%}) = \exp(\hat{\delta}^{DP})SE(\hat{\delta}^{DP})$.

Table VIII: Data Package Effects: PII Fraction

	Dependent Variable: PII Fraction			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.002 (0.007)		-0.005 (0.011)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		-0.013 (0.028)		-0.020 (0.023)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.003 (0.008)		-0.004 (0.013)
Multinational			0.021 (0.020)	0.022 (0.020)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.423	0.423	0.424	0.424

** $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$*

Note: Standard errors are clustered at the country level. PII fraction is the number of records in a data package considered PII divided by the total number of records in the data package.

Table IX: Data Package Effects: Number of Data Types

	Dependent Variable: Number of Unique Data Types			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.420*		0.392	
	(0.225)		(0.245)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.263		0.232
		(0.586)		(0.585)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.437*		0.409
		(0.260)		(0.279)
Multinational			0.086	0.087
			(0.231)	(0.231)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.280	0.280	0.280	0.280

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level.

Figures

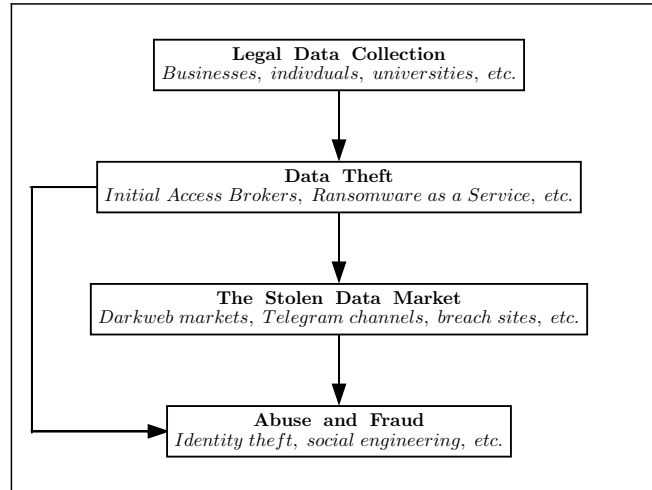


FIGURE I
The Stolen Data Supply Chain

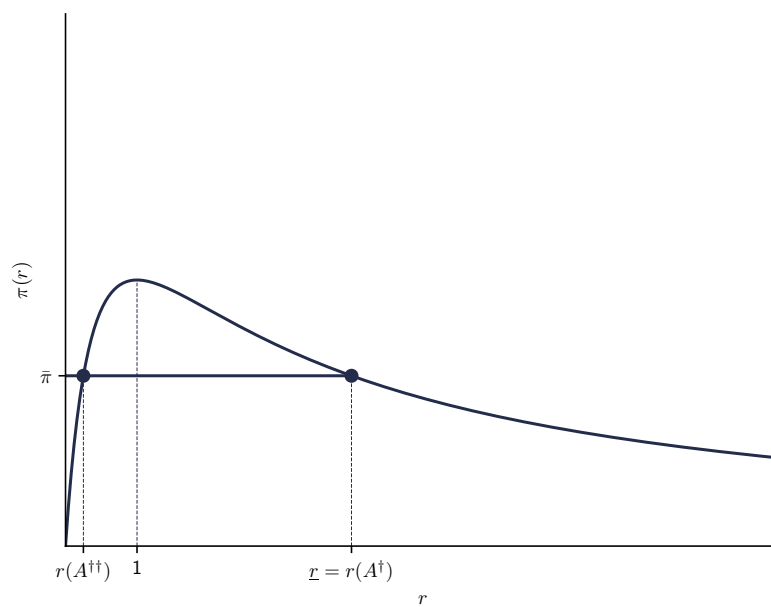


FIGURE II
 $\pi(r)$: Breach Probability as a Function of r

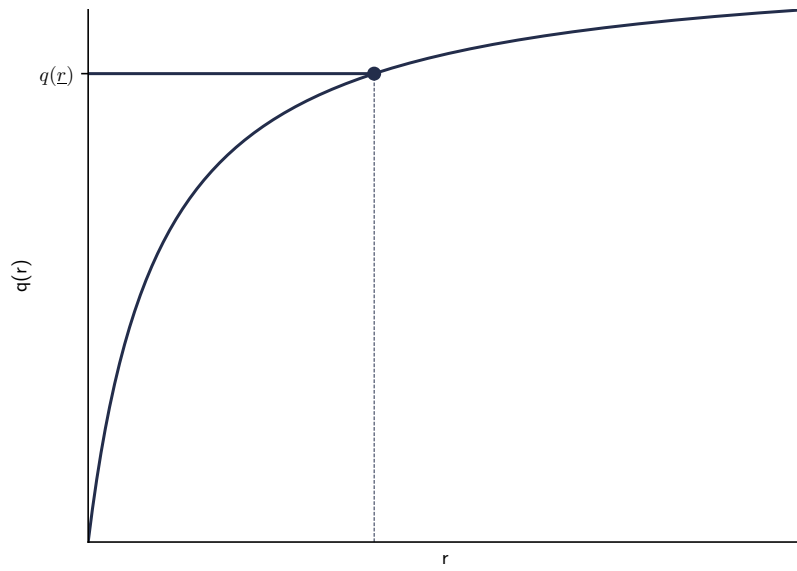


FIGURE III
 $q(r)$: Portion of Legitimate Data Packages as a Function of r

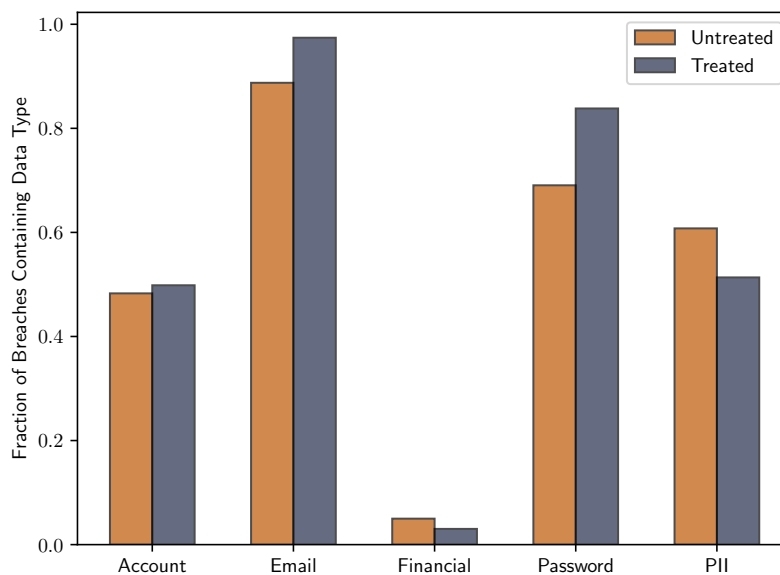
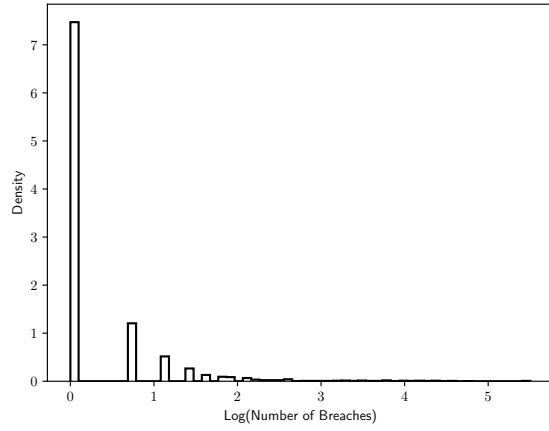


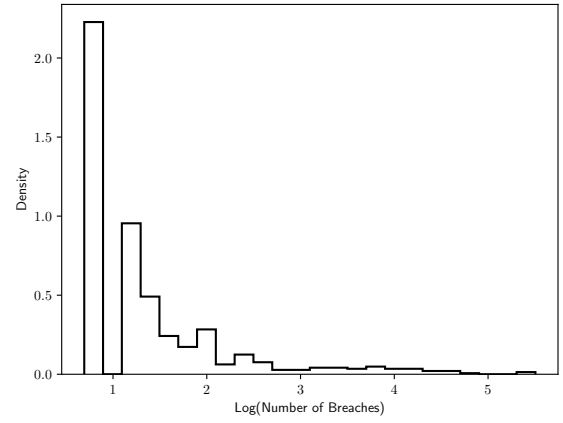
FIGURE IV

Fraction of Data Packages Containing Each Data Type

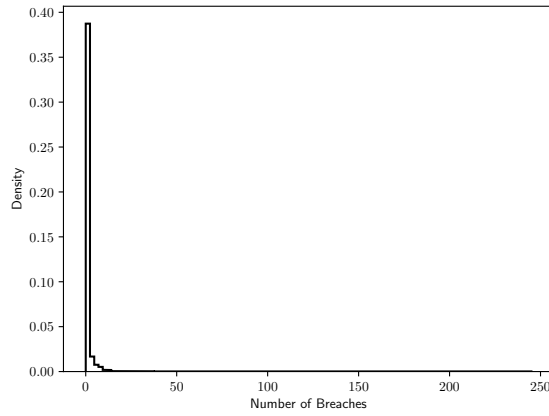
Notes: The password category includes both individual passwords themselves, and information related to passwords, such as the salt used to help obscure them. Financial information includes bank, credit card, and loan data. Treated refers to all data packages originating in the EU, untreated to those originating outside the EU.



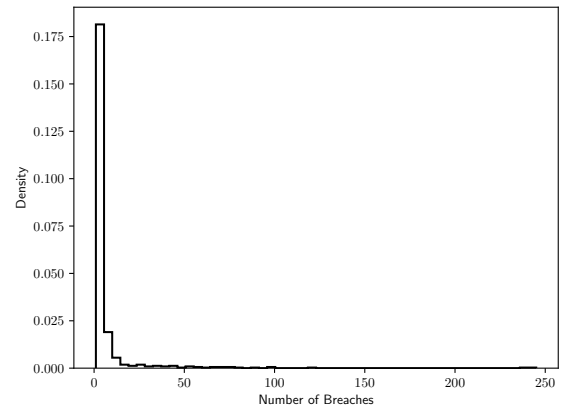
(A) Logged Using All Observations



(B) Logged Using Only Positive Observations

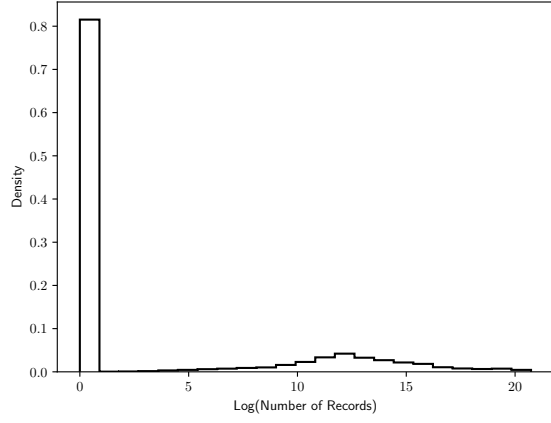


(C) In Levels Using All Observations

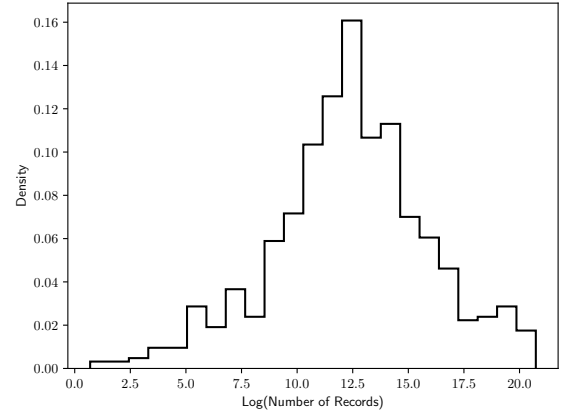


(D) In Levels Using Only Positive Observations

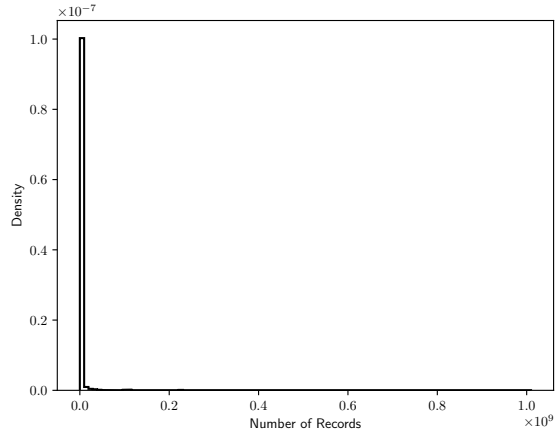
FIGURE V
Distribution of the Aggregate Number of Data Breaches Per Country by Quarter



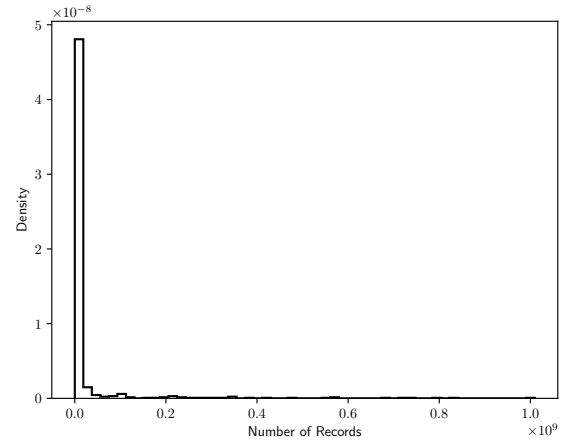
(A) Distribution of Log(Number of Records) Using All Observations



(B) Distribution of Log(Number of Records) Using Only Positive Observations

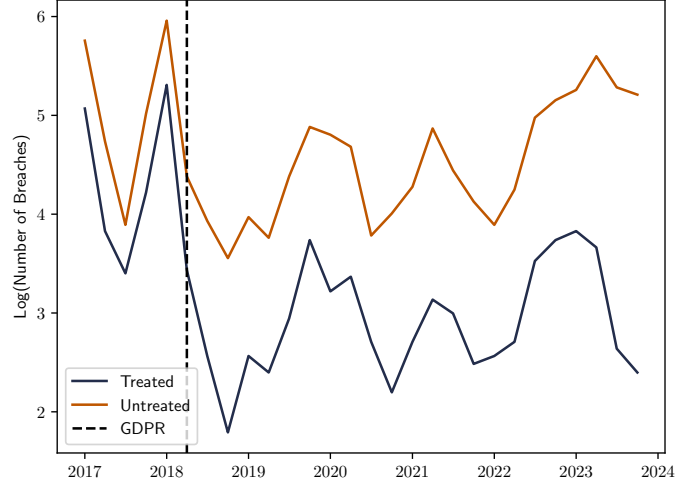


(C) Distribution of Number of Records Using All Observations

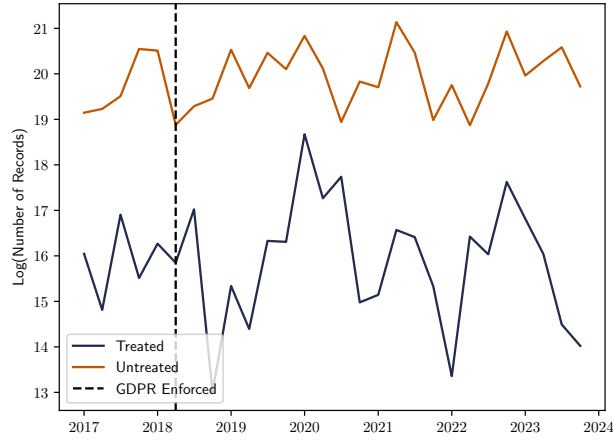


(D) Distribution of Number of Records Using Only Positive Observations

FIGURE VI
Distribution of the Aggregate Number of Records Per Country by Quarter



(A) Number of Data Breaches



(B) Number of Records Online

FIGURE VII
Number of Breaches and Records Online Time Series

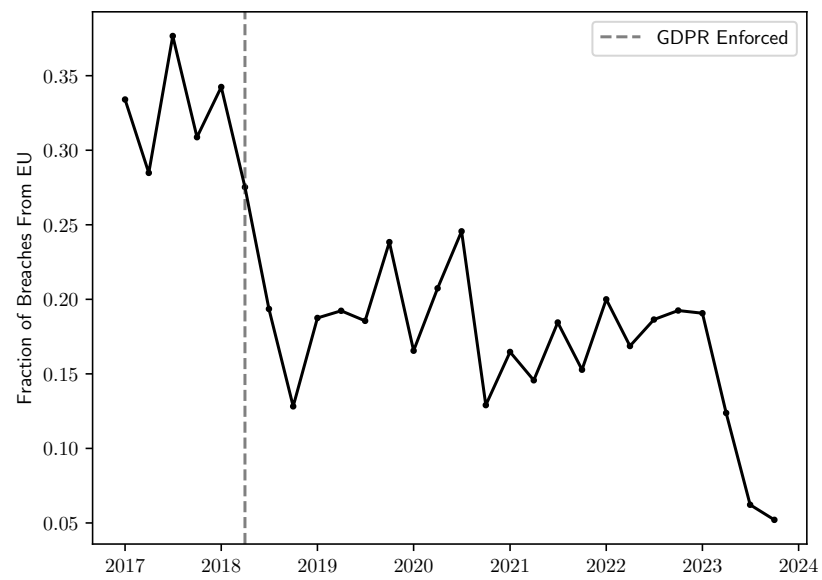
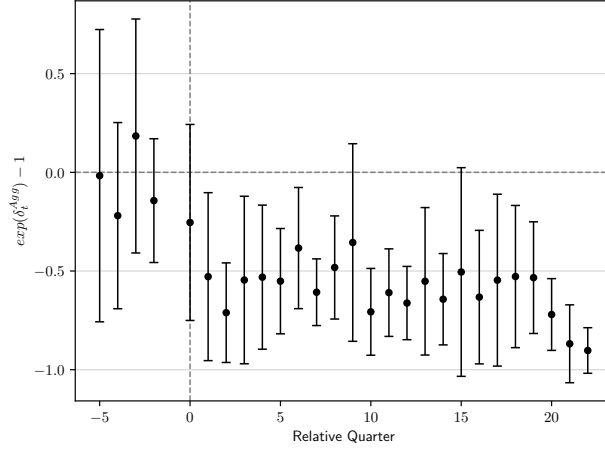
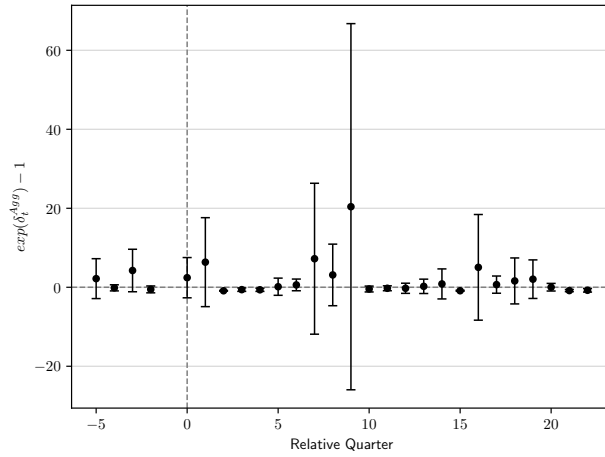


FIGURE VIII
Fraction of Breaches That Originate in EU



(A) Number of Data Breaches



(B) Number of Records Available

FIGURE IX
Aggregate Effect Event Studies

Note: The figures present estimates of the δ_{it}^{Agg} coefficients in equation 11 converted to percentage changes using $\exp(\delta_{it}^{Agg}) - 1$. The bars are the 95 percent confidence intervals with standard errors clustered at the country level. Period $t = -1$, the first quarter of 2018, is normalized to be zero.

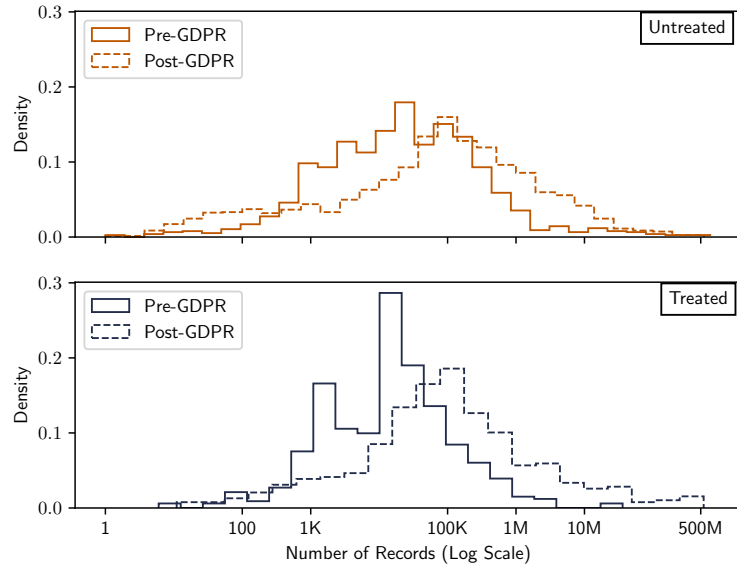


FIGURE X
Number of Records Per Data Package Density

Note: The distributions shown are of the number of records in each data package. This differs from figure V, which is the distribution of the aggregate number of records observed for each country in all periods.

THE EFFECTS OF PRIVACY REGULATION ON THE MARKET FOR STOLEN DATA ONLINE APPENDIX

A. Data

A.A UK Survey Data

The UK survey data referenced throughout the paper are from the United Kingdom Cyber Security Breach Survey: Combined Dataset, 2016-2022 (Department for Digital, Culture, Media and Sport, [2022](#)). I accessed the data through the UK Data Services online portal on March 20, 2023.

Only the 2018 and 2019 surveys asked respondents whether they made any changes in response to the GDPR. The survey asked about the types of changes made as well, which I have combined into five groups: human changes (e.g., staff training and hiring), technical changes (e.g., updated system configurations and increased spending on security), policy changes (e.g., conducting more audits and changing who has admin rights), third-party changes (e.g., changing IT service providers), and other changes (e.g., changing the nature of the business).

For figure [XI](#), an organization was considered breached if they reported a ransomware or other malware infection; hacking of bank accounts; phishing attacks; unauthorized file access; or any other breach or attack.

A.B Breach Data

The individual breach data obtained for this study contains many more data package observations than are included in the final paper. Observations were dropped for one of three

reasons. First, any breaches that could not be attributed to an organization or country were removed. Second, any data package that was discovered during a breach of a breach-indexing website, or similar “breach of breaches” was dropped. These breaches are of websites and other platforms that bundle access to credentials leaked in other breaches to their users. Essentially, the data being leaked in those breaches had itself been stolen from its original owner. What makes these observations unusable is the lack of a clear date when the data were originally stolen. The observed date is of the larger breach, but it is unknown when the smaller breaches that comprise the breach occurred. Finally, data packages that appeared online prior to 2017 were removed. As briefly discussed when the panel data was described, the organization collecting these data was founded in 2016. Dropping these early breaches allows for the possibility that the breaches collected prior to that founding were meaningfully different from those that were collected later.

B. Defining Personally Identifiable Information

From a legal standpoint, there are three commonly used definitions of “personally identifiable information” (Schwartz and Solove, 2011). The tautological definition used in the Video Privacy Protection Act says that PII is information which identifies a person. The non-public information approach used in the Gramm-Leach-Bliley Act defines PII as non-public personal information. Finally, the specific-types approach explicitly lists the types of data that are considered PII. I borrow from all three approaches.

In the data I am able to observe the specific types of records in the data packages. I classify data as PII if it reveals location, financial, contact, user account, or personal information. Account information covers emails, usernames, and passwords. Personal information includes political and religious views, sexual orientation, and aspects of a person’s home life such as if they have children or pets. As most of the data packages included emails and passwords (figures IV and XII), this makes the fraction of records in a data package that are

PII fairly close to one. As part of my robustness checks, I repeat the data package analysis of the effect of the GDPR on the fraction of records in a data package that are PII using an alternative definition that removes emails and passwords. I find that this did not change the main result that the GDPR had no effect on the portion of records in a breach that are PII (table XXXIII).

C. Descriptive Information

Tables XI-XIV report unconditional differences in means between various data package groups.

Table XI compares treated and untreated data packages across the full sample. There are statistically significant differences between the two in the fraction of records that are PII, and the number of unique data types. Although they are statistically significant, they are not particularly meaningful. Given that both types have close to 70 percent PII, a 4 percentage point difference is not particularly large. And the difference in number of unique data types is less than one, making them effectively the same from an interpretation perspective.

Data packages that became available before and after the GDPR are then compared in table XII. The data packages get significantly larger after the GDPR in terms of both the number of records and the number of unique data types. As shown in tables XIII and XIV, which compare the packages pre- and post-GDPR for the control and treated groups, respectively, this effect is seen in both, though it is much larger in the treated group. This is consistent with the findings that expected data package size significantly increased after the GDPR, and the theory that attackers may have shifted their efforts towards larger targets.

C.A Dropped Observations

The observations dropped from the final datasets because either their country of origin or breach date could not be determined tend to contain fewer records than those included in

the study (figure XIII). There are three periods where a large number of breaches were dropped: the first and second quarters of 2018, and the fourth quarter of 2020. In each of these periods there was a data breach whose contents were an amalgamation of data from many other smaller breaches. The 2020 breach specifically, known as the Cit0Day breach, was a collection of over 23,000 breached websites bundled together. The Cit0Day website collected each of those smaller breaches and offered access to the information they contained for a fee. These observations are dropped because it is not possible to identify when these smaller breaches occurred. It is possible they were breaches that occurred years prior to the larger breach, or right before. Figure XIV plots the number of breaches and records included and excluded from the final sample over time.

D. Results

Additional results from alternative specifications of the estimated models are presented here.

D.A Extensive Margin Effects

On the extensive margin, I separately estimate equation IV.A for small and large countries. The former are countries with populations below the median population in 2018, the latter countries with populations above the median population in 2018. Results are in tables XV and XVI.

D.B Aggregate Effects

To test the robustness of my aggregate effect estimates, I re-estimate each aggregate effect after removing a treated country from the data. For each removed country, the estimate stays well within the 95 percent confidence interval of the estimate with the full sample (figures XV and XVI).

Next, I use different methods to construct the panel. Brazil and China each adopted

their own data privacy laws near the end of the study period. Removing them from the sample slightly reduces the estimated effect on the number of breaches, though it is still significant. As before, there are no statistically significant effects on the number of records available (table [XVII](#)).

Table [XVIII](#) shows the aggregate results when I exclude data from after the first quarter of 2020 to avoid any pandemic effects. This significantly reduces the number of post-treatment observations. The change in the number of records remains insignificant and in number of data breaches significant, but the long-run effect in the latter case does change. The short-run effects on both outcomes are identical to using the full panel, which is unsurprising since observations in the pre-treatment period and short-run all remain in this new panel. The only change is in the long-run estimates, where the reduction in number of data breaches shrank, though is still statistically significant. In this shorter panel there are only three long-run periods: the third and fourth quarters of 2019 and the first quarter of 2020. These results suggest that the long-run effect grows as time goes on.

Table [XIX](#) shows the quantity results when I exclude data packages originating in multinational organizations from the panel. Whether an organization is a multinational is determined in one of two ways. First, if their website is hosted in more than one country, they are considered multinational. Or, they are considered multinational if their website and organizational information, such as privacy policies, discuss having customers or users in more than one country. The argument for excluding these effects is that multinational organizations may be partially treated. The GDPR applies to data specifically from EU residents. A multinational organization would therefore have to comply if they have any users in the EU, but it is not clear whether they would change their data collection and protection practices for all their users, or just those in the EU.

When multinational breaches are excluded, there is actually a long-run increase in the number of records available after the GDPR. The effect on the number of data breaches is roughly equivalent to the one found in the main specification.

Each of the previous tests left the definition of the outcome variables unchanged and were estimated with the same Poisson regression as in the main paper. Tables [XX-XXV](#) test changes in the outcome variable definition, the effect of adding covariates to the equation, and using three other models to derive estimates.

First, I estimate the effect using a linear model, rather than a Poisson model:

$$Y_{it} = \gamma_s + \tau_t + \delta D_i \times Post-GDPR_t + \epsilon_{it} \quad (17)$$

where each term is defined as before. In addition to using a linear model, I use two log-like transformations of the outcome variable, $\log(Y_{it} + 1)$ and the inverse hyperbolic sine (IHS) function $\ln(Y_{it} + \sqrt{Y_{it}^2 + 1})$. These transformations are necessary, rather than just using $\log(Y_{it})$ because there are a number of periods in which countries have no breaches. Using these transformations significantly changes the results from the Poisson model. For the number of records, both the $\log(Y_{it} + 1)$ and IHS transformation give large and statistically significant negative estimates of the treatment effect, unlike the Poisson which showed no change. I believe this is due to a significant extensive margin effect. J. Chen and Roth, [2023](#) and Mullahy and Norton, [2024](#) both discuss how, when there are mass points at zero, log-like transformations may greatly influence the estimated coefficients.

As discussed in the main body of the paper, there are significant and negative extensive margin effects (table [IV](#)). This is likely the source of the discrepancies in effect sizes between the models and the primary reason for using the Poisson model over the linear models with a log-like transformation.

Without the log-like transformation, when it is estimated in levels, the linear model produces results that are in line with, though interpreted differently than, the Poisson model. Specifically, I still find no effect on the number of records and a significant negative effect on the number of data breaches (columns 7 and 8 of each table).

Next, I estimate the models using various measures to account for population size. In tables [XXII](#) and [XXIII](#), I add population in millions as a covariate. It is not included

in the levels models because the outcomes are already scaled to be records/data packages per million. In all cases there is no significant change in the estimates and the population coefficient is insignificant.

In tables [XXIV](#) and [XXV](#), I change the outcome for the log-like transformation to also be number of records/data packages per million, and add a population offset to the Poisson model. This noticeably changes the magnitude of both log-like transformations in each outcome. As J. Chen and Roth, [2023](#) discuss, this is a reflection of the sensitivity of log-like transformations to the scale of the outcome variable when extensive margin effects are present. The offset in the Poisson model effectively changes the outcome to a rate, as in breaches per million. The estimates, however, are roughly the same as those in the model without the offset.

For my final model specification tests, I used alternative ways of controlling for population and added covariates. My results were largely unchanged in each case. In table [XXVI](#), I remove the population offset. Table [XXVII](#) weights the estimates using population rather than including an offset. In table [XXVIII](#), I again remove the population offset and opt instead for using per capita outcome variables. Finally, I split the sample into small and large countries in tables [XXIX](#) and [XXX](#), and include indicators for whether the observation is a small or large country in table [XXXI](#).¹⁷

A shortcoming of my data is that many variables that would be reasonable to include as covariates, such as the fraction of people with internet access, are not consistently observed for every country. Rather than drop observations and unbalance the panel to account for this, the only covariate I add to the model is GDP per capita. As shown in table [XXXII](#), this does not have a significant effect on my effect estimates.

¹⁷Small or large in this context means above or below the median population in 2018.

D.C Data Package Effects

Estimates of the change in PII fraction using a slightly different definition of PII are in table [XXXIII](#). Under this definition, I remove emails and passwords from PII. I find no significant change, as is the case using the original definition.

As previously discussed, data packages from periods prior to January 2017 were excluded from the main dataset. Tables [XXXV-XXXVII](#) report the results using the full sample, including those early breaches. In all cases, the signs of the estimated coefficients remain the same. The magnitude of the increase in number of records is larger (comparing table [XXXV](#) to table VI), but the estimates are still within each other's standard errors. For the number of unique data types, using the full sample does result in a statistically significant increase, unlike the smaller sample. But the increase is still less than a single data type and therefore not economically meaningful.

Finally, I estimated extensive margin effects for each of the data types using the linear probability model

$$Positive_i = \gamma_i + \tau_t + \delta D_{it} + \varepsilon_{it}$$

where $Positive_i$ is one if the data package contains a positive amount of that data; γ_i and τ_t are country and quarter fixed effects, respectively; and D_{it} is an indicator for whether the data package is treated.

There is a short-run increase in the probability of a data package containing email addresses and password information, but neither is maintained into the long run. Long term, the only data type showing a significant change is account information, which saw an eight percent increase in the likelihood that it is in a data package (table [XXXVIII](#)).

Online Appendix Tables

Table X: UK Cyber Security Breach Survey Dates and Sample

Survey Year	Sample Size	Survey Period
2016	1,008 businesses	November 30, 2015 – February 5, 2016
2017	1,523 businesses	October 24, 2016 – January 11, 2017
2018	1,519 businesses, 569 charities	October 9, 2017 – December 14, 2017
2019	1,566 businesses, 514 charities	October 10, 2018 – December 20, 2018
2020	1,348 businesses, 337 charities	October 9, 2019 – December 23, 2019
2021	1,419 businesses, 487 charities, 378 educational institutions	October 12, 2020 – January 21, 2021
2022	1,243 businesses, 424 charities, 490 educational institutions	September 20, 2021 – January 21, 2022

Note: Survey data and documentation can be accessed via the UK Data Service digital catalogue.

Table XI: Data Package Means: Treated vs. Untreated

	Means		Overall Mean N=4,394	Differences
	0 N=3,467	1 N=927		Treated - Untreated
Number of Records	3,309,228 (465,094.606)	4,422,933 (1,128,570.651)	3,544,186 (437,434.656)	1,113,705 (1,220,649.297)
PII Fraction	0.700 (0.003)	0.665 (0.006)	0.693 (0.003)	-0.035*** (0.007)
# of Data Types	6.363 (0.089)	5.688 (0.166)	6.220 (0.079)	-0.674*** (0.188)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table XII: Data Package Means: Pre-GDPR vs. Post-GDPR

	Means		Overall Mean N=4,394	Differences
	0 N=1,621	1 N=2,773		Post - Pre
Number of Records	1,598,365 (465,366.681)	4,681,646 (636,601.781)	3,544,186 (437,434.656)	3,083,281*** (788,560.699)
PII Fraction	0.551 (0.003)	0.776 (0.003)	0.693 (0.003)	0.225*** (0.004)
# of Data Types	3.163 (0.069)	8.007 (0.104)	6.220 (0.079)	4.844*** (0.125)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table XIII: Data Package Means: Pre- vs. Post-GDPR, Untreated

	Means		Overall Mean N=3,467	Differences
	0 N=1,175	1 N=2,292		Post - Pre
Number of Records	2,123,526 (640,456.742)	3,917,082 (621,924.394)	3,309,228 (465,094.606)	1,793,556** (892,734.445)
PII Fraction	0.552 (0.004)	0.776 (0.004)	0.700 (0.003)	0.224*** (0.005)
# of Data Types	3.279 (0.090)	7.943 (0.113)	6.363 (0.089)	4.664*** (0.145)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table XIV: Data Package Means: Pre- vs. Post-GDPR, Treated

	Means		Overall Mean N=927	Differences
	0 N=446	1 N=481		Post - Pre
Number of Records	214,813 (92,543.690)	8,324,849 (2,159,205.555)	4,422,933 (1,128,570.651)	8,110,036*** (2,161,187.859)
PII Fraction	0.547 (0.004)	0.775 (0.008)	0.665 (0.006)	0.228*** (0.009)
# of Data Types	2.859 (0.080)	8.312 (0.259)	5.688 (0.166)	5.453*** (0.271)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table XV: Extensive Margin Effects: Small Countries

	Dependent Variable: Positive Number of Breaches	
	(1)	(2)
Post $\times$ Treatment	-0.213*** (0.049)	
SR $\times$ Treatment		-0.211*** (0.064)
LR $\times$ Treatment		-0.213*** (0.049)
Observations	1,372	1,372
R^2	0.323	0.323
Period Fixed Effects	Y	Y
Country Fixed Effects	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Small countries are defined as those with a population below the median in 2018.

Table XVI: Extensive Margin Effects: Large Countries

	Dependent Variable: Positive Number of Breaches	
	(1)	(2)
Post $\times$ Treatment	-0.166*** (0.060)	
SR $\times$ Treatment		-0.100 (0.088)
LR $\times$ Treatment		-0.181*** (0.058)
Observations	1,344	1,344
R^2	0.509	0.509
Period Fixed Effects	Y	Y
Country Fixed Effects	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Large countries are defined as those with a population above the median in 2018.

Table XVII: Aggregate Effects: Dropping Brazil and China

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.830*** (0.229)		0.451 (0.461)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.793*** (0.300)		-0.283 (0.577)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.834*** (0.243)		0.546 (0.455)
$\hat{\delta}^{Agg\%}$	-0.564 (0.100)		0.570 (0.723)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.548 (0.136)		-0.246 (0.435)
$\hat{\delta}_{LR}^{Agg\%}$		-0.566 (0.106)		0.726 (0.785)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,660	2,660	2,660	2,660
Pseudo R^2	0.811	0.811	0.885	0.885

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Brazil and China have been removed from the panel. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XVIII: Aggregate Effects: Excluding COVID Years

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.649*** (0.182)		-0.218 (0.572)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.794*** (0.299)		-0.222 (0.592)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.541*** (0.152)		-0.212 (0.576)
$\hat{\delta}^{Agg\%}$	-0.478 (0.095)		-0.196 (0.460)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.548 (0.135)		-0.199 (0.474)
$\hat{\delta}_{LR}^{Agg\%}$		-0.418 (0.089)		-0.191 (0.466)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	852	852	852	852
Pseudo R^2	0.849	0.849	0.885	0.885

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. All periods after the first quarter of 2020 are excluded from the panel. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XIX: Aggregate Effects: Excluding Multinational Organizations

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.915*** (0.280)		0.652 (0.434)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.818*** (0.288)		-0.722 (0.704)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.923*** (0.300)		0.803* (0.443)
$\hat{\delta}^{Agg\%}$	-0.599 (0.112)		0.919 (0.833)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.559 (0.127)		-0.514 (0.342)
$\hat{\delta}_{LR}^{Agg\%}$		-0.603 (0.119)		1.232 (0.989)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,632	2,632	2,632	2,632
Pseudo R^2	0.784	0.784	0.824	0.825

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Data packages originating from multinational organizations are excluded from the panel construction. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\delta^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XX: Alternative Models: Number of Data Breaches

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-0.935*** (0.266)		-0.392*** (0.081)		-0.488*** (0.098)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.797*** (0.297)		-0.350*** (0.106)		-0.439*** (0.129)		-0.123*** (0.032)
LR $\times$ Treatment		-0.947*** (0.284)		-0.401*** (0.077)		-0.500*** (0.093)		-0.105** (0.044)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.692	0.692	0.682	0.682	0.102	0.102
Pseudo R^2	0.793	0.793						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of data breaches per million. Unlike the main specification, the Poisson model does not include a population offset.

Table XXI: Alternative Models: Number of Records

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.338 (0.429)		-2.225*** (0.391)		-2.361*** (0.414)		0.949 (20.561)	
SR $\times$ Treatment		-0.223 (0.590)		-1.673*** (0.556)		-1.780*** (0.587)		-0.718 (17.996)
LR $\times$ Treatment		0.402 (0.429)		-2.352*** (0.393)		-2.495*** (0.417)		1.333 (22.551)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.545	0.545	0.542	0.543	0.182	0.182
Pseudo R^2	0.847	0.847						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of records per thousand. Unlike the main specification, the Poisson model does not include a population offset.

Table XXII: Alternative Models with Covariates: Number of Data Breaches

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-1.090*** (0.308)		-0.390*** (0.083)		-0.486*** (0.099)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.847*** (0.322)		-0.353*** (0.106)		-0.443*** (0.129)		-0.123*** (0.032)
LR $\times$ Treatment		-1.117*** (0.324)		-0.400*** (0.078)		-0.497*** (0.094)		-0.105** (0.044)
GDP Per Capita	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		
Population	-0.004 (0.006)	-0.005 (0.006)	0.004 (0.003)	0.004 (0.003)	0.006 (0.004)	0.005 (0.004)		
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648	2,648	2,648	2,716	2,716
R^2			0.696	0.696	0.686	0.687	0.102	0.102
Pseudo R^2	0.797	0.797						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of data breaches per million. Annual population data is provided by the World Bank. Unlike the main specification, the Poisson model does not include a population offset.

Table XXIII: Alternative Models with Covariates: Number of Records

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.478 (0.509)		-2.212*** (0.386)		-2.350*** (0.409)		0.949 (20.561)	
SR $\times$ Treatment		-0.151 (0.618)		-1.721*** (0.550)		-1.831*** (0.581)		-0.718 (17.996)
LR $\times$ Treatment		0.564 (0.520)		-2.343*** (0.389)		-2.489*** (0.413)		1.333 (22.551)
GDP Per Capita	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		
Population	0.015 (0.015)	0.016 (0.015)	0.020 (0.022)	0.017 (0.022)	0.021 (0.023)	0.017 (0.023)		
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648	2,648	2,648	2,716	2,716
R^2			0.551	0.551	0.549	0.549	0.182	0.182
Pseudo R^2	0.847	0.848						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of records per thousand. Annual population data is provided by the World Bank. Unlike the main specification, the Poisson model does not include a population offset.

Table XXIV: Alternative Models: Number of Data Breaches Scaled by Population

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-0.927*** (0.266)		-0.090*** (0.020)		-0.110*** (0.026)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.794*** (0.297)		-0.093*** (0.020)		-0.112*** (0.025)		-0.123*** (0.032)
LR $\times$ Treatment		-0.939*** (0.284)		-0.090*** (0.020)		-0.109*** (0.026)		-0.105** (0.044)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.251	0.251	0.230	0.230	0.102	0.102
Pseudo R^2	0.792	0.793						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. The dependent variable is the number of breaches per million, except in the Poisson model, where a log(population) offset is used instead. Annual population data is provided by the World Bank.

Table XXV: Alternative Models: Number of Records Scaled by Population

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.342 (0.429)		-0.493*** (0.120)		-0.599*** (0.139)		0.949 (20.561)	
SR $\times$ Treatment		-0.221 (0.590)		-0.412*** (0.152)		-0.502*** (0.178)		-0.718 (17.996)
LR $\times$ Treatment		0.407 (0.429)		-0.511*** (0.124)		-0.621*** (0.143)		1.333 (22.551)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.447	0.447	0.450	0.450	0.182	0.182
Pseudo R^2	0.847	0.847						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. The dependent variable is the number of records per thousand, except in the Poisson model, where a log(population) offset is used instead. Annual population data is provided by the World Bank.

Table XXVI: Aggregate Effects: No Offset

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.935*** (0.266)		0.338 (0.429)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.797*** (0.297)		-0.223 (0.590)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.947*** (0.284)		0.402 (0.429)
$\hat{\delta}^{Agg\%}$	-0.607 (0.104)		0.402 (0.602)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.549 (0.134)		-0.200 (0.472)
$\hat{\delta}_{LR}^{Agg\%}$		-0.612 (0.110)		0.495 (0.642)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	0.793	0.793	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. No population offset is used. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXVII: Aggregate Effects: Weighted Estimation

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.113*** (0.384)		-0.007 (0.473)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.933*** (0.212)		-0.476 (0.558)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.128*** (0.411)		0.040 (0.497)
$\hat{\delta}^{Agg\%}$	-0.671 (0.126)		-0.007 (0.469)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.607 (0.083)		-0.379 (0.347)
$\hat{\delta}_{LR}^{Agg\%}$		-0.676 (0.133)		0.041 (0.517)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	1.426	1.426	1.031	1.031

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. The population offset is removed and observations are instead weighted by population. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXVIII: Aggregate Effects: Per Capita Outcomes

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.810* (0.447)		0.506 (0.392)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.285 (0.449)		-0.294 (0.709)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.840* (0.467)		0.578 (0.400)
$\hat{\delta}^{Agg\%}$	-0.555 (0.199)		0.658 (0.650)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.248 (0.338)		-0.255 (0.528)
$\hat{\delta}_{LR}^{Agg\%}$		-0.568 (0.201)		0.783 (0.713)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	0.071	0.071	0.473	0.474

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXIX: Aggregate Effects: Small Countries

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.400*** (0.427)		0.120 (0.436)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.343 (0.674)		-2.253*** (0.626)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.454*** (0.426)		0.262 (0.430)
$\hat{\delta}^{Agg\%}$	-0.753 (0.105)		0.127 (0.491)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.291 (0.478)		-0.895 (0.066)
$\hat{\delta}_{LR}^{Agg\%}$		-0.766 (0.099)		0.300 (0.559)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	1,372	1,372	1,372	1,372
Pseudo R^2	0.473	0.474	0.746	0.751

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are limited to countries with populations below the median population level in 2018. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXX: Aggregate Effects: Large Countries

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.863*** (0.283)		0.290 (0.498)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.742** (0.328)		-0.089 (0.617)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.874*** (0.300)		0.338 (0.504)
$\hat{\delta}^{Agg\%}$	-0.578 (0.120)		0.336 (0.665)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.524 (0.156)		-0.085 (0.565)
$\hat{\delta}_{LR}^{Agg\%}$		-0.583 (0.125)		0.402 (0.707)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	1,344	1,344	1,344	1,344
Pseudo R^2	0.803	0.803	0.827	0.827

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are limited to countries with populations larger than the median population in 2018. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXXI: Aggregate Effects: Size Indicators

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Above Median Pop. x Post	-0.337 (0.360)		-0.528 (0.411)	
Above Median Pop. x SR		0.813* (0.450)		-0.606 (0.584)
Above Median Pop. x LR		-0.334 (0.391)		-0.461 (0.360)
Post x Treatment	-1.399*** (0.423)		0.111 (0.435)	
SR x Treatment		-0.323 (0.674)		-2.313*** (0.603)
LR x Treatment		-1.337** (0.540)		-0.031 (0.458)
Above Median Pop. x Post x Treatment	0.536 (0.507)		0.178 (0.657)	
Above Median Pop. x SR x Treatment		-0.477 (0.745)		2.265*** (0.855)
Above Median Pop. x LR x Treatment		0.273 (0.558)		0.503 (0.839)
Observations	2,716	2,648	2,716	2,648
Pseudo R^2	0.793	0.797	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are unweighted.

Table XXXII: Aggregate Effects: With Covariates

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.051*** (0.285)		0.392 (0.471)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.834*** (0.320)		-0.190 (0.596)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.072*** (0.298)		0.465 (0.482)
GDP Per Capita	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\hat{\delta}^{Agg\%}$	-0.650 (0.100)		0.480 (0.697)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.566 (0.139)		-0.173 (0.493)
$\hat{\delta}_{LR}^{Agg\%}$		-0.658 (0.102)		0.591 (0.768)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648
Pseudo R^2	0.796	0.797	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are unweighted. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table XXXIII: Data Package Effects: PII Fraction - Excluding Emails and Passwords

	Dependent Variable: PII Fraction			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.004 (0.020)		-0.013 (0.014)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.014 (0.042)		-0.004 (0.042)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.003 (0.022)		-0.014 (0.016)
Multinational			0.051*** (0.019)	0.051*** (0.019)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.422	0.422	0.423	0.423

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. PII definition excludes emails and passwords.

Table XXXIV: Data Package Effects: Number of PII Records - Excluding Emails and Passwords

	Dependent Variable: Log(Number of PII Records)			
	(1)	(2)	(3)	(4)
Post x Treatment	0.948 (0.638)		0.364 (0.390)	
SR x Treatment		1.846** (0.824)		1.383** (0.680)
LR x Treatment		0.897 (0.591)		0.369 (0.402)
Multinational			1.809*** (0.346)	1.741*** (0.337)
$\hat{\delta}$	1.579 (1.647)		0.439 (0.562)	
$\hat{\delta}^{SR}$		5.337 (5.224)		2.987 (2.710)
$\hat{\delta}^{LR}$		1.451 (1.448)		0.447 (0.582)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.382	0.383	0.386	0.386

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. PII definition excludes emails and passwords.

Table XXXV: Data Package Effects: Number of Records—Full Sample

	Dependent Variable: Log(Number of Records)			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.956** (0.400)		0.563** (0.254)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.657 (0.462)		0.221 (0.297)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.989** (0.405)		0.599** (0.266)
Multinational			1.424*** (0.308)	1.426*** (0.308)
$\hat{\delta}^{DP\%}$	1.602 (1.041)		0.756 (0.445)	
$\hat{\delta}_{SR}^{DP\%}$		0.929 (0.891)		0.248 (0.371)
$\hat{\delta}_{LR}^{DP\%}$		1.687 (1.089)		0.821 (0.484)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.280	0.280	0.288	0.289

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table XXXVI: Data Package Effects: PII Fraction—Full Sample

	Dependent Variable: PII Fraction			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	-0.003 (0.007)		-0.008 (0.009)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		-0.019 (0.030)		-0.025 (0.025)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		-0.001 (0.007)		-0.007 (0.011)
Multinational			0.020 (0.015)	0.020 (0.015)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.402	0.402	0.403	0.403

** $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$*

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table XXXVII: Data Package Effects: Number of Data Types—Full Sample

	Dependent Variable: Number of Unique Data Types			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.524** (0.249)		0.527** (0.243)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.263 (0.549)		0.266 (0.576)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.552* (0.291)		0.554* (0.280)
Multinational			-0.011 (0.207)	-0.010 (0.207)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.277	0.277	0.277	0.277

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table XXXVIII: Data Types Extensive Margin Effects

	Account		Email		Financial		Passwords		PII	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post $\times$ Treatment	0.085** (0.037)		0.033 (0.034)		-0.011 (0.012)		0.033 (0.061)		0.014 (0.027)	
SR $\times$ Treatment		0.087 (0.072)		0.061*** (0.016)		0.011 (0.027)		0.135*** (0.030)		-0.028 (0.065)
LR $\times$ Treatment		0.085* (0.044)		0.030 (0.040)		-0.014 (0.013)		0.023 (0.069)		0.019 (0.031)
Observations	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394
R^2	0.275	0.275	0.378	0.378	0.067	0.067	0.358	0.359	0.477	0.477
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. The extensive margin is estimated using a linear probability model. The dependent variable is an indicator for whether the data package contains data of each type. PII in columns 9 and 10 does not include emails or passwords.

Online Appendix Figures

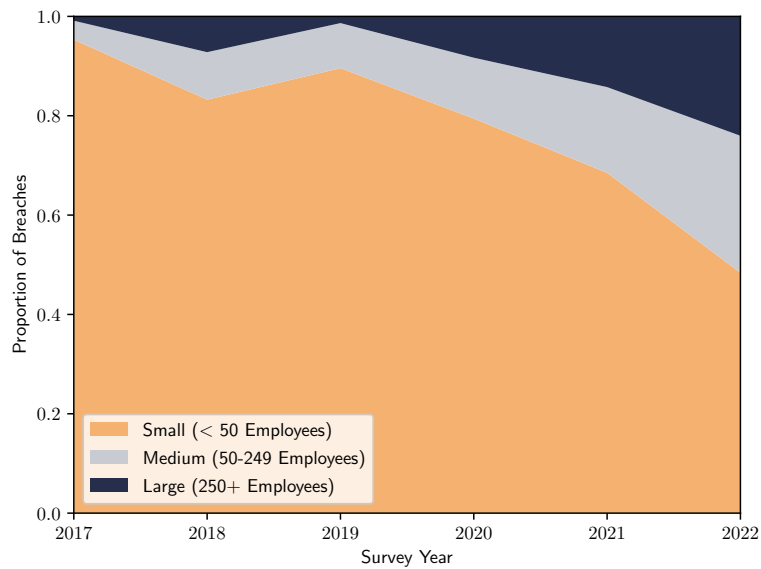
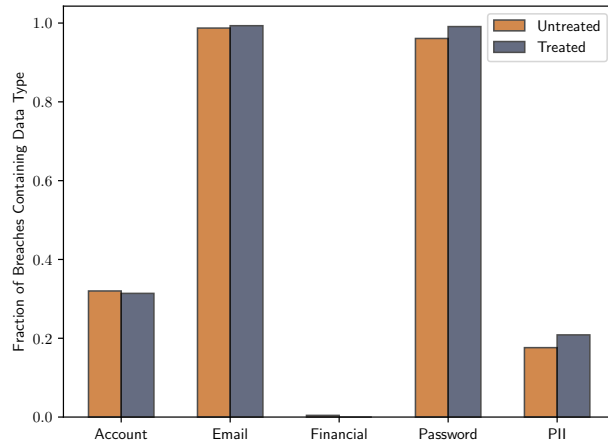
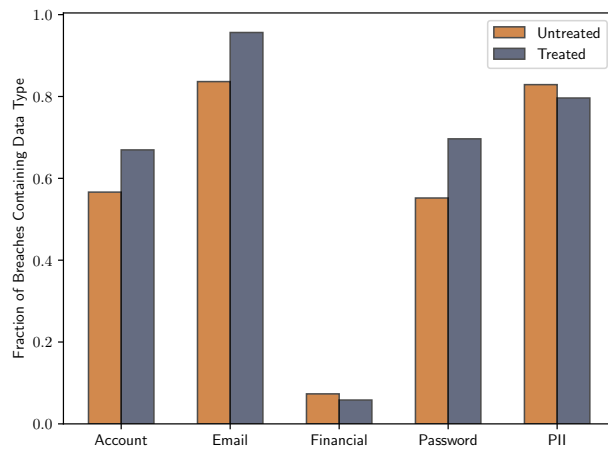


FIGURE XI
Portion of Data Breaches by Size

Note: This figure shows the proportion of *Source:* Department for Digital, Culture, Media and Sport (2022), author's calculations.



(A) Pre-GDPR



(B) Post-GDPR

FIGURE XII

Fraction of Data Packages Containing Each Data Type, Pre- and Post-GDPR

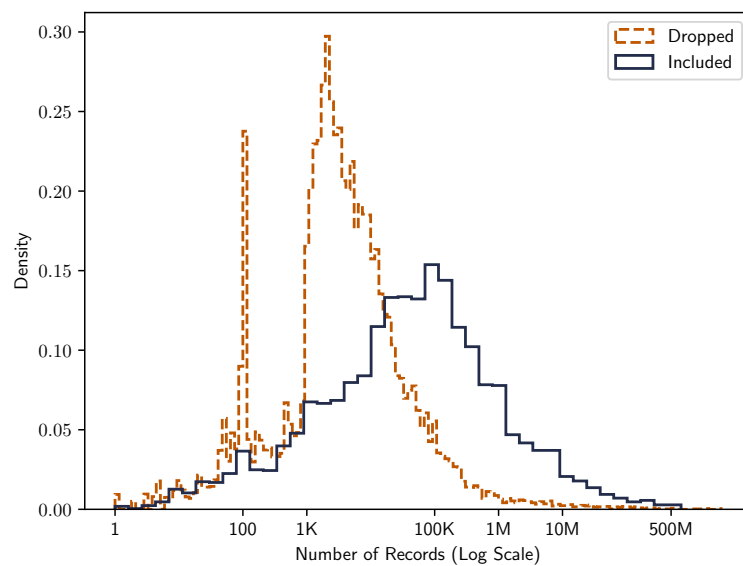
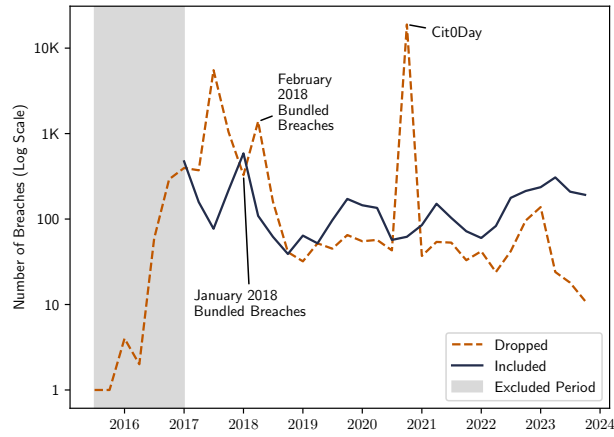
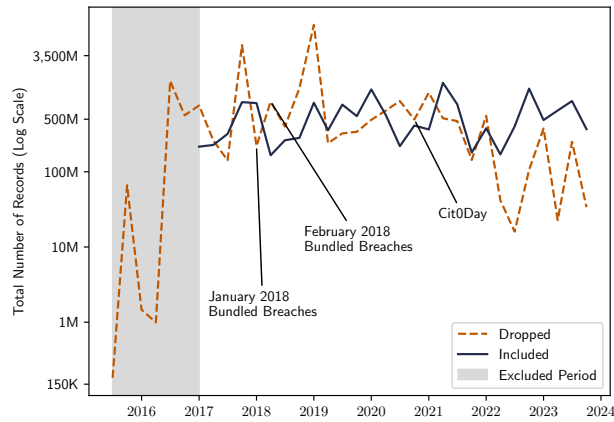


FIGURE XIII
Distribution of Records Per Breach: Dropped vs. Included

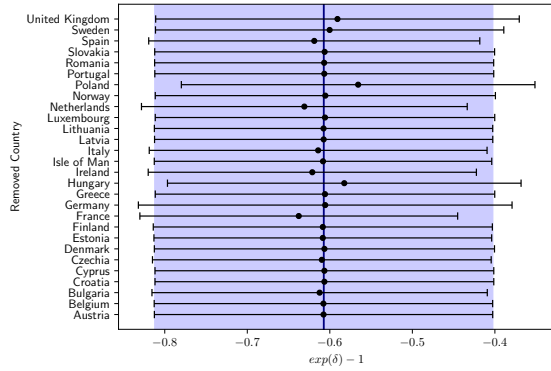


(A) Number of Breaches

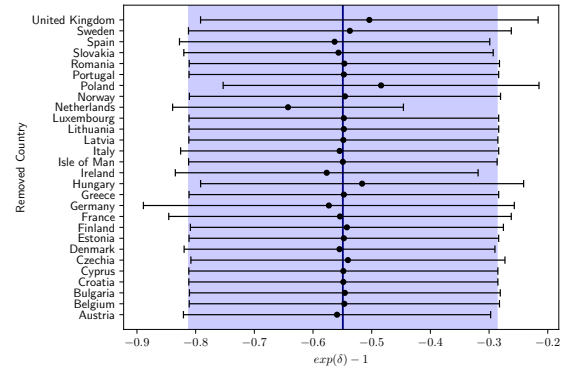


(B) Number of Records

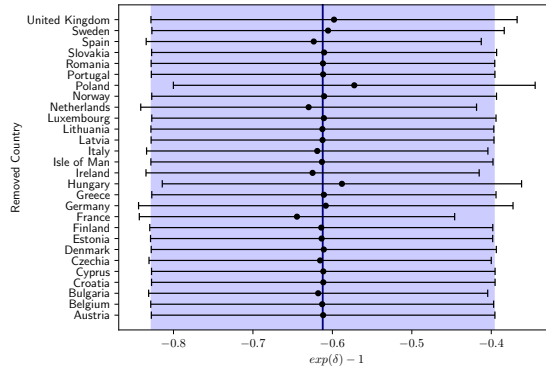
FIGURE XIV
Comparison of Dropped and Included Breaches Over Time



(A) Overall Effect



(B) Short Run Effect

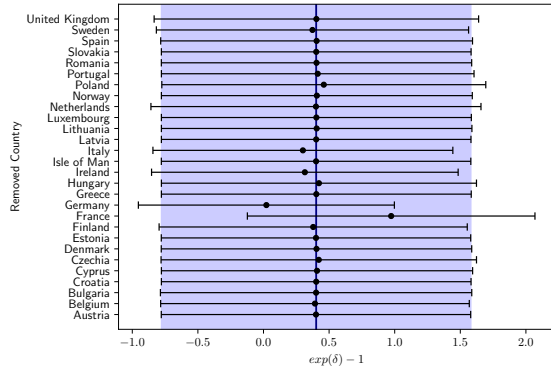


(C) Long Run Effect

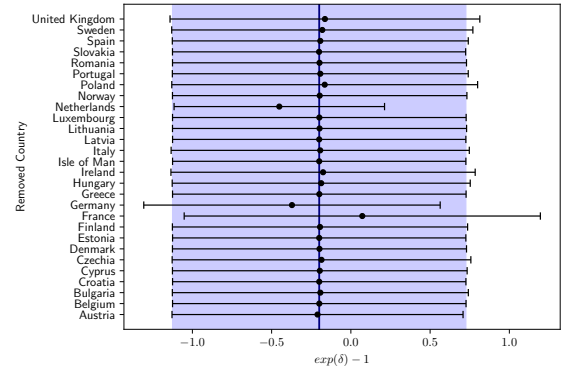
FIGURE XV

Number of Data Breaches Effects Removing Countries

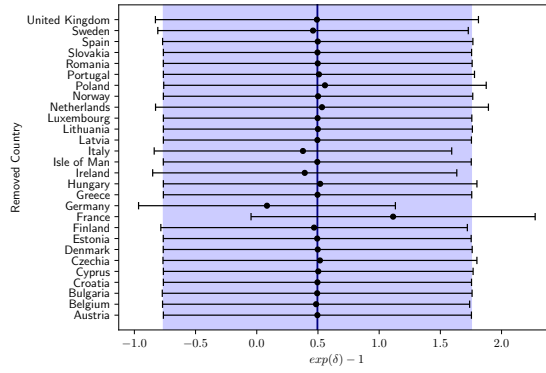
Note: Each point represents the estimated effect on number of data breaches after removing observations from the specified country. The whiskers are the 95 percent confidence interval. The solid line is the point estimate including all countries, and the shaded area is the 95 percent confidence interval around that point.



(A) Overall Effect



(B) Short Run Effect



(C) Long Run Effect

FIGURE XVI

Number of Records Effects Removing Countries

Note: Each point represents the estimated effect on number of records after removing observations from the specified country. The whiskers are the 95 percent confidence interval. The solid line is the point estimate including all countries, and the shaded area is the 95 percent confidence interval around that point.